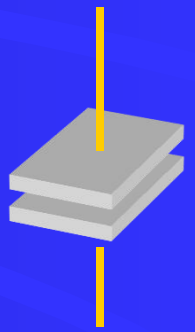
第5章 动态电路分析

5.1 动态元件

5.1.1 电容元件

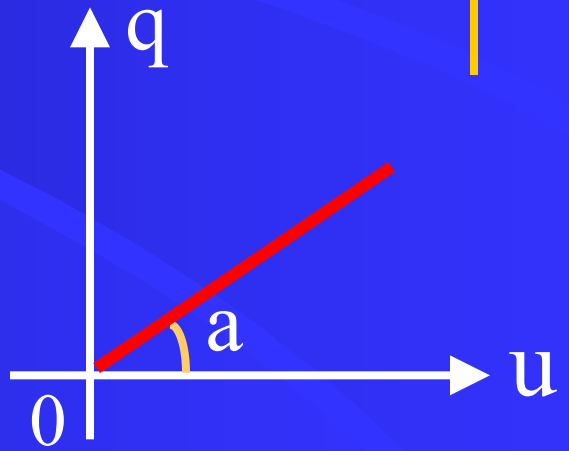
1. 由两片平行的金属极板、中间以绝缘介质隔开所构成的器件。

2. 电容是一种贮能元件。



电容的定义：

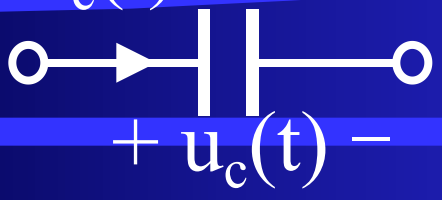
一个二端元件在任何时刻t， 它的电荷q(t)同它的端电压u(t)之间的关系，用q-u平面上的一条曲线来确定。



电容量：C=q/u=tan(a)

电容的符号和单位：

符号：$i_c(t)$ C



$1F=10^6\ \mu F=10^{12}\ PF$

单位：C =q(库仑)/u(伏特) =F(法拉）

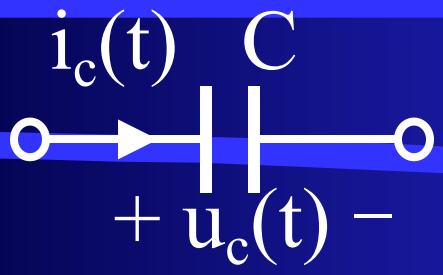
电容的额定工作电压：

如：6.3V−10μf

此电容工作在6.3V以上的电路中，它就会被击穿。

电容电流的大小及方向：

关联方向



由电容定义：$C = q/u_c \quad \therefore q = Cu_c$

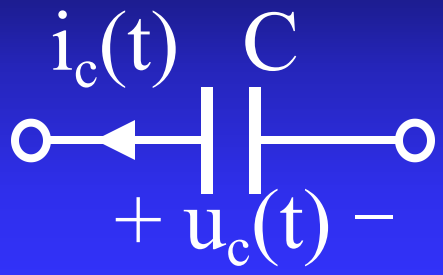
由电流定义：

$$i_c(t) = \frac{dq}{dt} = \frac{dCu_c(t)}{dt} = C\frac{du_c(t)}{dt}$$

(1) 某一时刻的电流取决于该时刻电压的变化率。

(2) 电容电压变化越快,电流就越大。

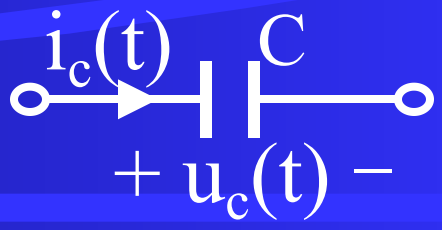
非关联方向



$$i_c(t) = -C\frac{du_c(t)}{dt}$$

电容中的电压：

$$\because i_c(t) = C\frac{du_c(t)}{dt}$$



$$\therefore u_c(t) = \frac{1}{C}\int_{-\infty}^{t} i(\xi)d\xi = \frac{1}{C}\int_{-\infty}^{t_0} i(\xi)d\xi + \frac{1}{C}\int_{t_0}^{t} i(\xi)d\xi$$

$$= u_c(t_0) + \frac{1}{C}\int_{t_0}^{t} i(\xi)d\xi \qquad t \geq t_0$$

$u_c(t_0)$：t_0时的电压，初始时刻的电压

$\frac{1}{C}\int_{t_0}^{t} i(\xi)d\xi$：$t_0$以后的电压，激励作用的电压



电容电压的两个重要性质：

1. 电容电压有记忆　$u_c(t) = U + u_1(t)$

　　U：初始电压

2. 电容电压不能突变具有连续性　$u_c(t_{0-}) = u_c(t_{0+})$

电容的储能：

由功率的定义：$p = \frac{dw}{dt}, \therefore dw = pdt$

$i(\xi) = c\frac{du(\xi)}{d\xi}$

$$\therefore w_c(t) = \int_{t_0}^{t} pdt = \int_{t_0}^{t} u(\xi) \times i(\xi) d\xi = \int_{t_0}^{t} Cu(\xi) \times du(\xi)$$

利用换元积分：

令 $u = u(\xi), \therefore \begin{cases} \xi = t_0 \text{时}, & u = u(t_0) \\ \xi = t \text{时}, & u = u(t) \end{cases}$

上式：$= C\int_{u(t_0)}^{u(t)} u\,du = \frac{1}{2}Cu^2\Big|_{u(t_0)}^{u(t)}$

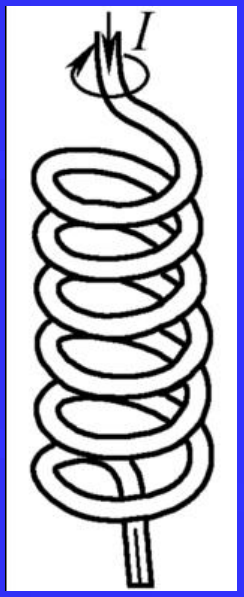
$$= \frac{1}{2}Cu^2(t) - \frac{1}{2}Cu^2(t_0)$$

若$u(t_0)=0$

电容的储能：

$W_c=1/2Cu^2(t)$

5.1.2 电感元件

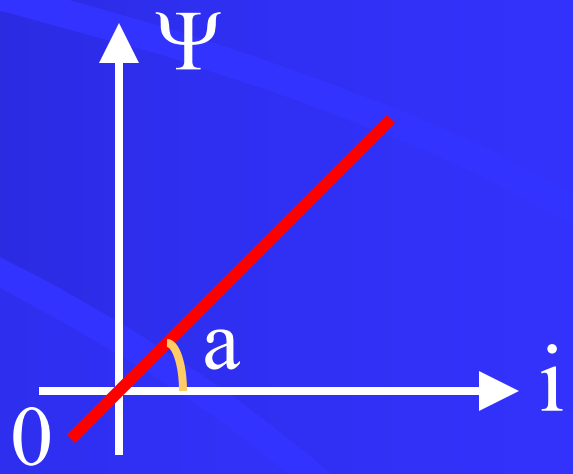


1. 实际电感器是用导线绕制的线圈。当导线中有电流流过时，在导线周围就会产生磁场，在线圈中就会产生感应电压。电感元件则是电感线圈的理想化模型。

2. 电感是一种贮能元件。

电感的定义：

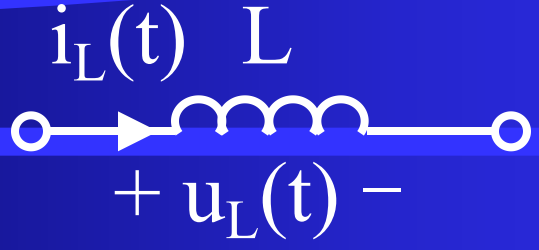
在任何时刻t，它的电流i(t)同它的磁链Ψ(t)之间的关系，用i-Ψ平面上的一条曲线来确定。



电感量:L= Ψ /i=tan(a)

电感的符号和单位：

符号：



单位：L= Ψ (韦伯)/i(电流) =H(亨利)

$$1\mathrm{H}=10^{3}\,\mathrm{mH}=10^{6}\,\mu\mathrm{H}$$

电感的额定工作电流：

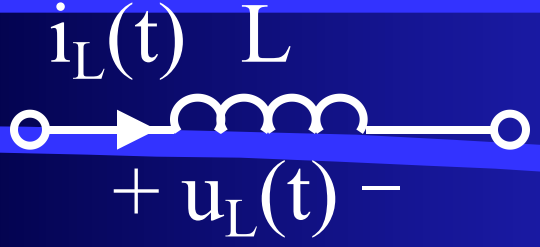
若超过额定工作电流,电感线圈产生机械变形,或烧毁。

实际线圈：



电感电压的大小和极性：

关联方向　　根据：　电磁感应定律：

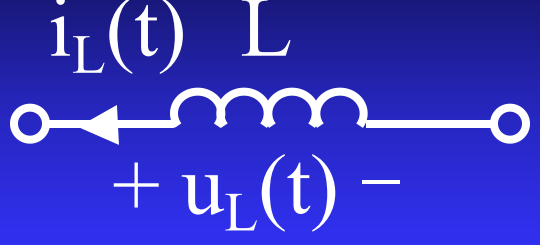


$$u_L(t)=\frac{d\psi(t)}{dt}=\frac{d\,Li_L(t)}{dt}=L\frac{di_L(t)}{dt}$$

(1) 某一时刻的电压取决于该时刻的电流变化率。

(2) 电流变化越快,电压就越大。

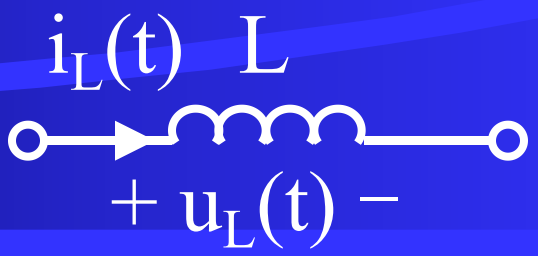
非关联方向



$$u_L(t)=-L\frac{di_L(t)}{dt}$$

电感中的电流：

$$\because u_L = L\frac{di}{dt}$$



$$\therefore i_L(t) = \frac{1}{L}\int_{-\infty}^{t} u_L(\xi)d\xi = \frac{1}{L}\int_{-\infty}^{t_0} u_L(\xi)d\xi + \frac{1}{L}\int_{t_0}^{t} u_L(\xi)d\xi$$

$$= i_L(t_0) + \frac{1}{L}\int_{t_0}^{t} u_L(\xi)d\xi \quad t \geq t_0$$

t_0时的电流　t_0以后的电流

初始时刻　　激励作用

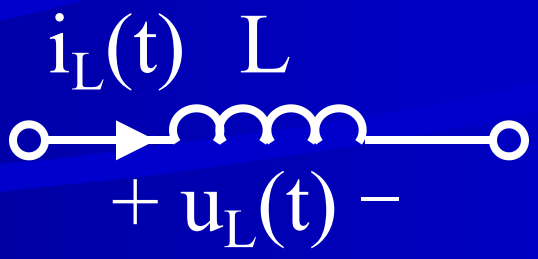
电感电流的两个重要性质：

1. 电感电流有记忆性　$i_L(t) = I + i_1(t)$

初始电流

2. 电感电流不能突变具有连续性　$i_L(t_{0-}) = i_L(t_{0+})$

电感的储能：



$$\because\ p=\frac{dw}{dt}$$

$$p(t)=u_L(t)\times i_L(t)$$

$$\therefore\ w_L(t)=\int_{t_0}^{t}pdt=\int_{t_0}^{t}u(\xi)\times i(\xi)d\xi$$

$$=\int_{t_0}^{t}L\frac{di(\xi)}{d\xi}\times i(\xi)d\xi$$

$\xi=t, dt=d\xi$

$t=i(\xi), dt=di(\xi)$

$$=\int_{t_0}^{t}L\,i(\xi)di(\xi)=L\int_{i(t_0)}^{i(t)}i\,di$$

$$=\frac{1}{2}Li^2\Big|_{i(t_0)}^{i(t)}=\frac{1}{2}Li^2(t)-\frac{1}{2}Li^2(t_0)$$

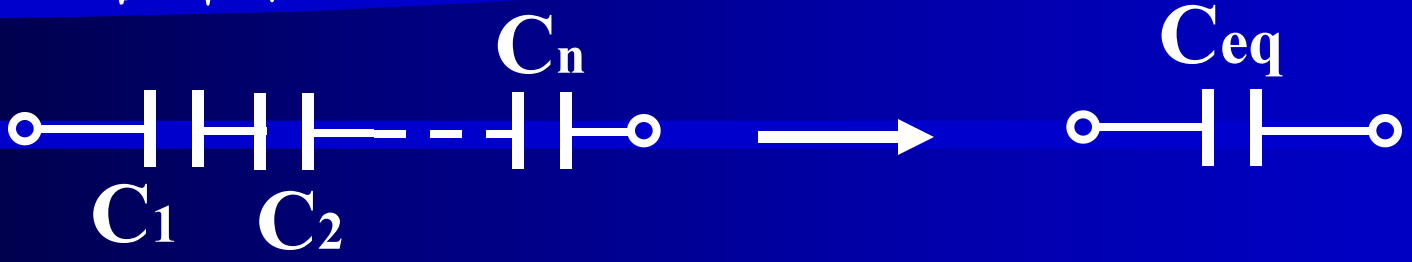
若$i(t_0)=0$

电感储能

$W_L=1/2Li^2(t)$

5.1.3 电容、电感元件的串联与并联

1. 电容串联:

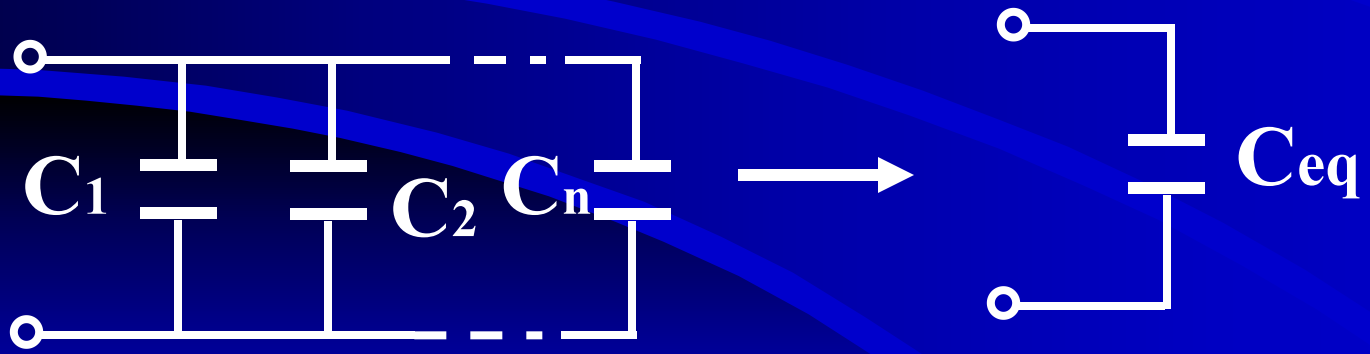


等效电容: $\frac{1}{C_{eq}}=\frac{1}{C_1}+\frac{1}{C_2}+\cdots+\frac{1}{C_n}$

若2个电容串联:

$$C_{eq}=\frac{C_1\cdot C_2}{C_1+C_2}$$

2. 电容并联:



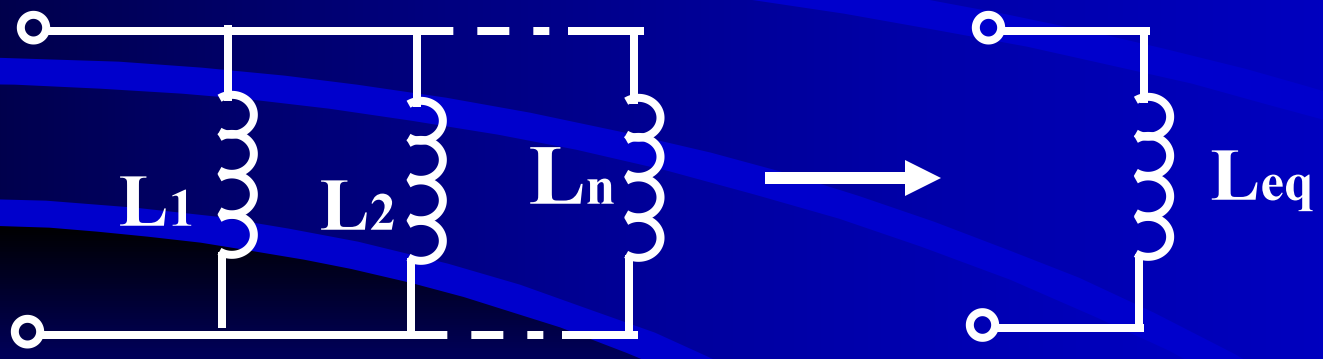
等效电容: $C_{eq}=C_1+C_2+\cdots+C_n$

3.电感串联



等效电感: $L_{eq} = L_1 + L_2 + \cdots + L_n$

4. 电感并联



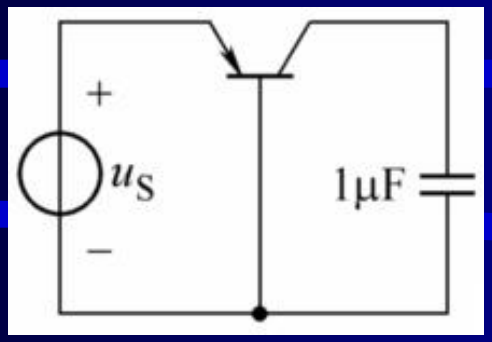
等效电感: $\dfrac{1}{L_{eq}} = \dfrac{1}{L_1} + \dfrac{1}{L_2} + \cdots + \dfrac{1}{L_n}$

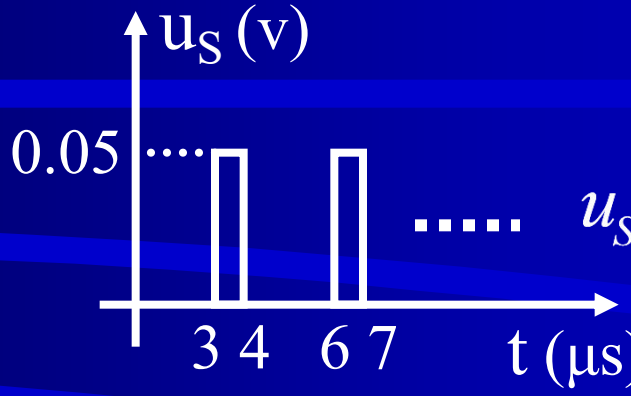
若2个电感并联:

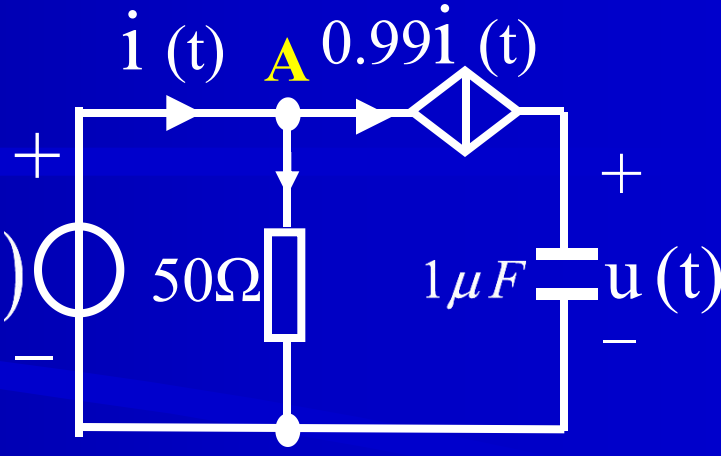
$$L_{eq} = \frac{L_1 \cdot L_2}{L_1 + L_2}$$

5.1.4 电容、电感元件的应用

脉冲记数器电路







脉冲电压源的脉冲宽度为1uS，脉冲幅度为0.05V。流经电容的电流为0.99i时，在该电流的作用下，电容的电压为

$$u(t)=\frac{1}{C}\int_{t_0}^{t}0.99i(t)dt$$

*A*点：由*KCL*:

$$i(t)=\frac{u_S(t)}{50}+0.99i(t)\quad\therefore i(t)=2u_S(t)$$

第一个脉冲作用：　$t_0 = 3\mu S \sim t = 4\mu S$

$\therefore i(t) = 2 \times 0.05 = 0.1\text{A}$

$$\therefore u(t) = \underbrace{u(3 \cdot 10^{-6})}_{0} + \frac{1}{10^{-6}} \int_{3 \cdot 10^{-6}}^{4 \cdot 10^{-6}} 0.99 \cdot 0.1 dt = 0.099\text{V}$$

第二个脉冲作用：　$t_0 = 6\mu S \sim t = 7\mu S$

$\therefore i(t) = 2 \times 0.05 = 0.1\text{A}$

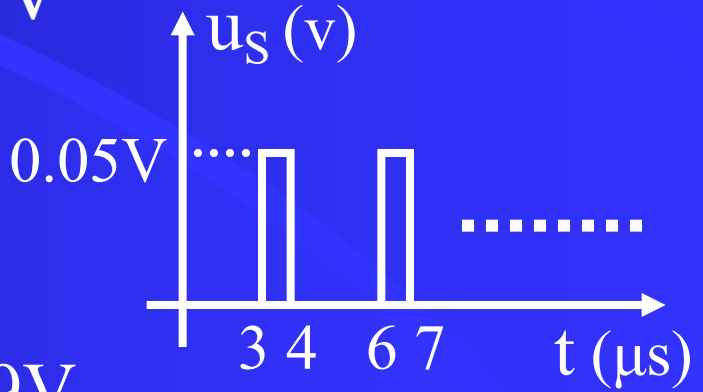
$$\therefore u(t) = u(6 \cdot 10^{-6}) + \frac{1}{10^{-6}} \int_{6 \cdot 10^{-6}}^{7 \cdot 10^{-6}} 0.99 \cdot 0.1 dt$$

$$= 0.099 + 0.099 = 0.198\text{V}$$

同理：第三个脉冲作用时，

u (t)又增加0.099V

当由100个脉冲时，输出电压为9.9V。



5.2 电路初始值的计算

5.2.1 换路定律

$$u_c(0_+) = u_c(0_-)$$

$$i_L(0_+) = i_L(0_-)$$

换路瞬间，若电容电流和电感电压为有限值，则电容的电压、电感的电流在该处连续，不会发生跃变。

5.2.2 初始值的确定

电容电压的初始值$u_C(0_+)$和电感电流的初始值$i_L(0_+)$均由原稳态电路$t=0_-$时刻的$u_C(0_-)$和$i_L(0_-)$来确定。

换路时其它电压、电流的初始值，如$i_C(0_+)$、$u_L(0_+)$、$u_R(0_+)$、$i_R(0_+)$等则可能发生跃变，必须在$t=0_+$时刻的等效电路中确定。

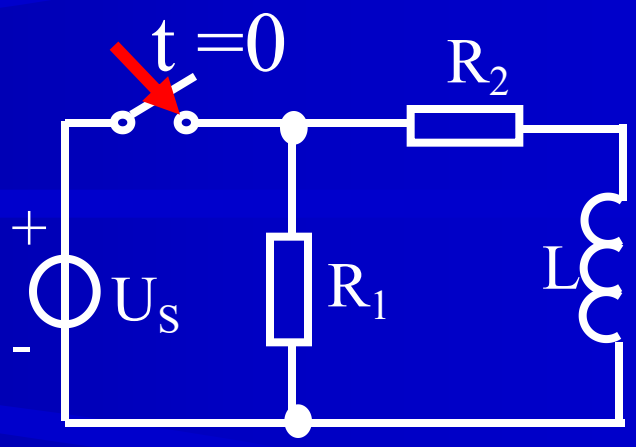
在$t=0_+$的等效电路中：

电容用一个电压值为$u_C(0_+)$的电压源代替，若$u_C(0_+)=0$时，电容用短路线代替。

电感用一个电流值为$i_L(0_+)$的电流源代替，若$i_L(0_+)=0$时，电感用开路代替。

其它元件如独立源、受控源和电阻等保持不变。

例1： 已知开关闭合前，L无贮能。求开关闭合后各电压、电流的初始值。



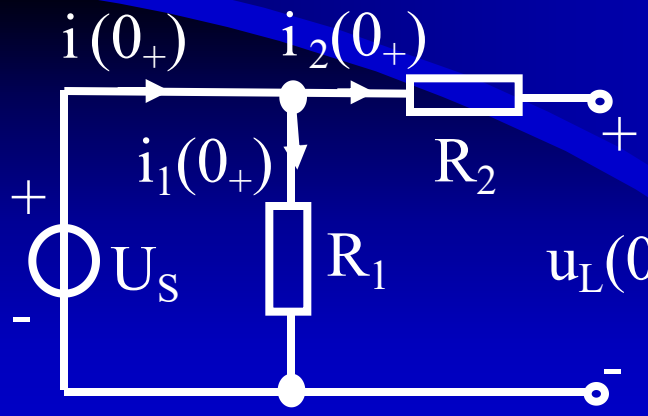
解： 已知开关闭合前为： $t = 0_-$

已知开关闭合后为： $t = 0_+$

并知 $i_L(0_-)=0$，∴ $i_L(0_+)= i_L(0_-)= 0$

∴在 $t = 0_+$ 的瞬间，L相当于开路。

∴ $t = 0_+$ 的等效电路为：



$i(0_+)=i_1(0_+)=U_S/R_1$

$i_2(0_+)=0$

$u_L(0_+)=i_1(0_+)\cdot R_1=U_S$

总结: 求初始值的步骤

1. 先求动态元件t =0$_-$时刻的值，再由换路定理求动态元件t =0$_+$时刻的值。

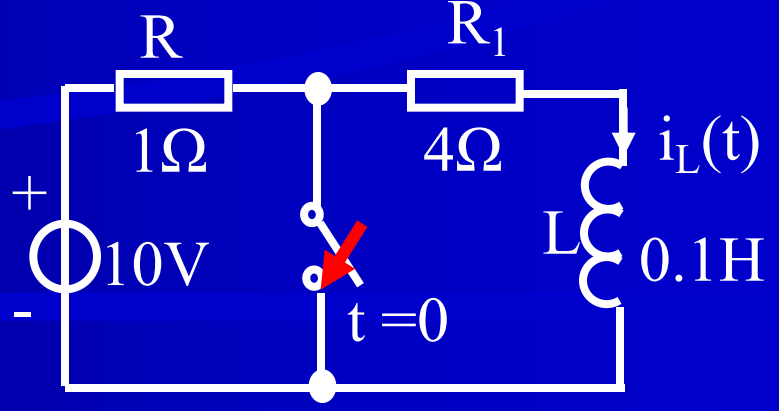
$$u_c(0_+) = u_c(0_-)$$

$$i_L(0_+) = i_L(0_-)$$

2. 作t =0$_+$时刻的等效电路。

3. 在t =0$_+$时刻的等效电路中求各量的初始值。

例2： 已知开关闭合前电路已达到稳定，求开关闭合后各电压、电流的初始值。



解： 1. 已知$t=0_-$时，电路达到稳定，

即$t\to\infty$ 时，L→短路。

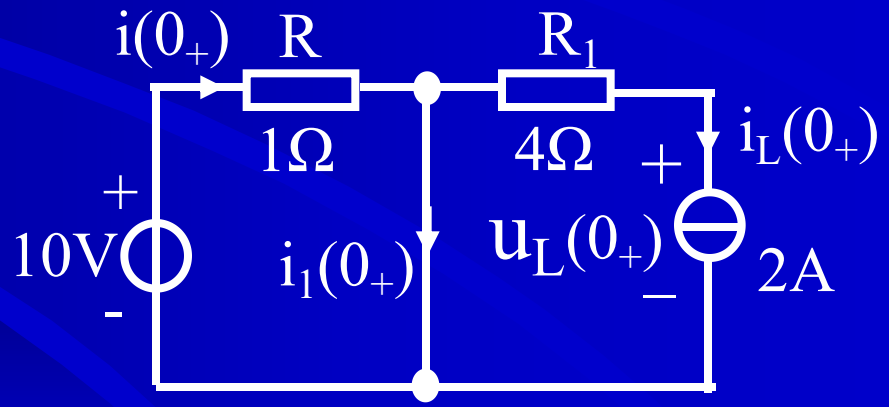
$$\therefore i_L(0_-)=\frac{10}{1+4}=2\text{A}$$

由 $i_L(0_+)=i_L(0_-)$

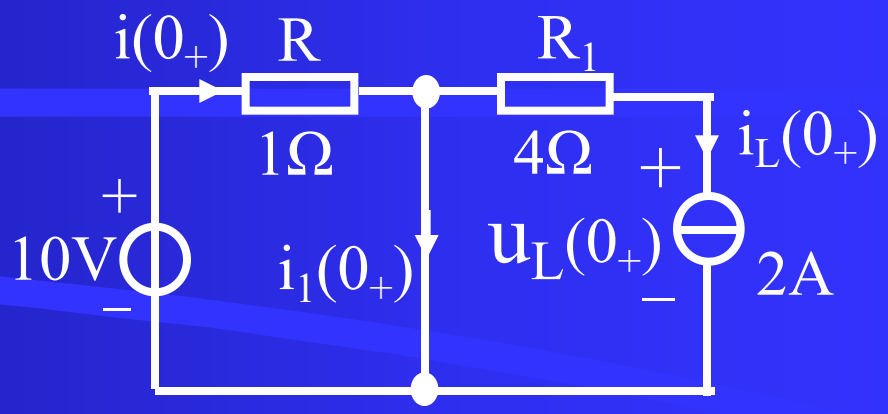
知 $i_L(0_+)=2\text{A}$

2. 作$t=0_+$时刻的等效电路。

（L→2A的电流源）



3. 根据t =0$_+$时的等效电路，求各量的初始值。



$$i(0_+)=\frac{10}{1}=10\text{A}$$

$$\because i(0_+)=i_1(0_+)+i_L(0_+)$$

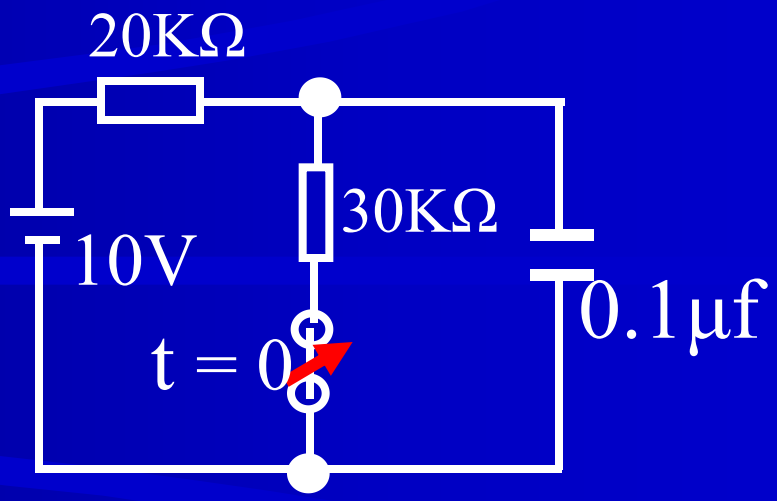
$$\therefore i_1(0_+)=i(0_+)-i_L(0_+)=10-2=8\text{A}$$

$$u_R(0_+)=i(0_+)\cdot R=10\cdot 1=10\text{V}$$

$$u_{R1}(0_+)=i_L(0_+)\cdot R_1=2\cdot 4=8\text{V}$$

$$u_L(0_+)=-u_{R1}=-8\text{V}$$

例3： 已知开关打开前电路已达到稳定，求开关打开后各电压、电流的初始值。



解： 1. $t=0_-$时，电路已达到稳定，即$t\to\infty$时，C→开路。

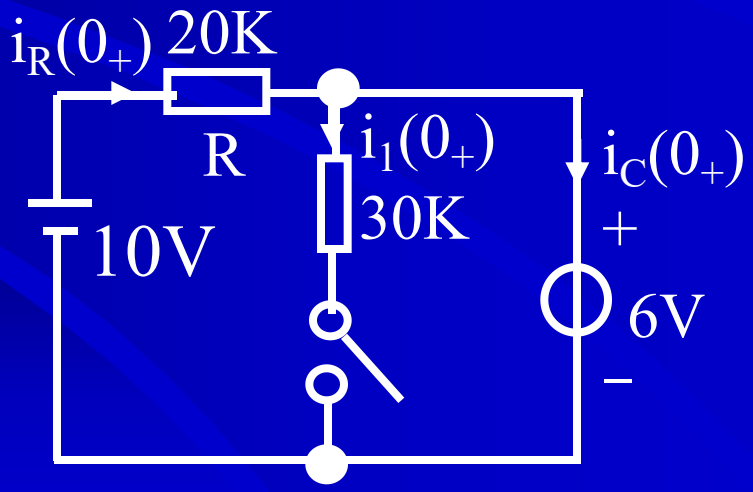
$$\therefore u_C(0_-)=\frac{30}{20+30}\cdot 10=6\text{V}$$

由$u_C(0_+)=u_C(0_-)$

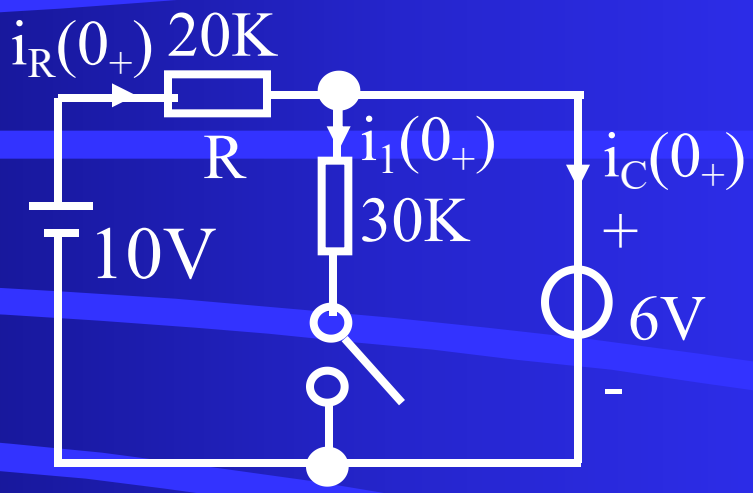
知$u_C(0_+)=6\text{V}$

2. 作$t=0_+$时刻的等效电路。

（C→6V的电压源）



3. 根据t =0$_+$时的等效电路，求各量的初始值。



$$i_R(0_+) = i_C(0_+) = \frac{10-6}{20} = 0.2\text{mA}$$

$$i_1(0_+) = 0$$

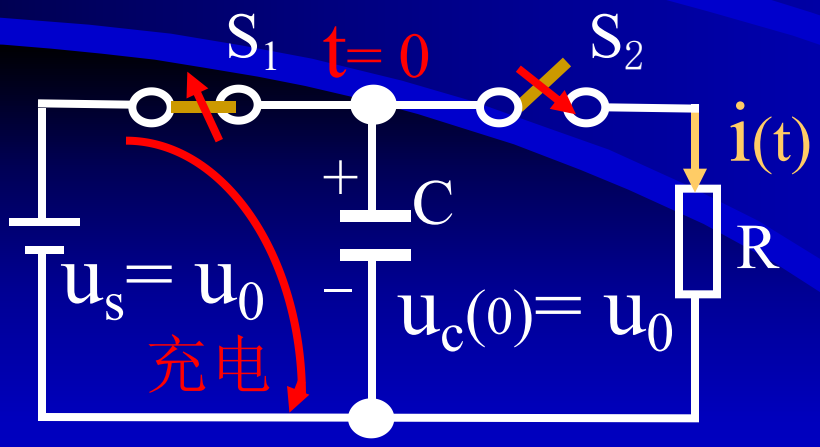
$$u_R(0_+) = i(0_+) \cdot R = 0.2 \cdot 20 = 4\text{V}$$

5.3 一阶电路分析

电路中只有一个动态元件，且电路方程用一阶微分方程来描述。

5.3.1 零输入响应

电路没有外加激励时，电路的响应由初始时刻电容中电场的储能，或电感中磁场的储能引起。



1. t <0时，C被充电为u_0。

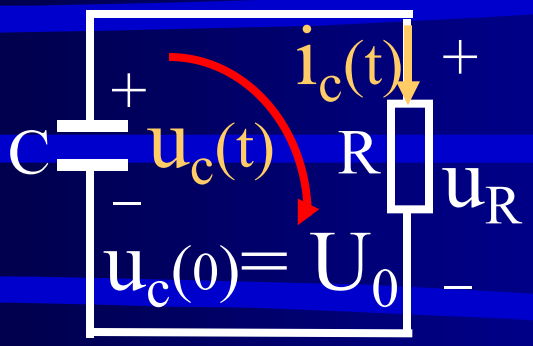
2. t = 0时，换路的瞬间，

因C上的电压不能突变，

$u_c(0_+)= u_0$， $i(t)= u_0/R$。

3. t ≥0时，换路后……。

1. RC电路的零输入响应：（由$u_c(0_+)$产生的响应）



由KVL: $u_R - u_c = 0 \qquad t \geq 0$

由元件的性质：$\begin{cases} u_R = R \times i_C \\ i_C = -C\dfrac{du_c}{dt} \end{cases}$ 代入上式

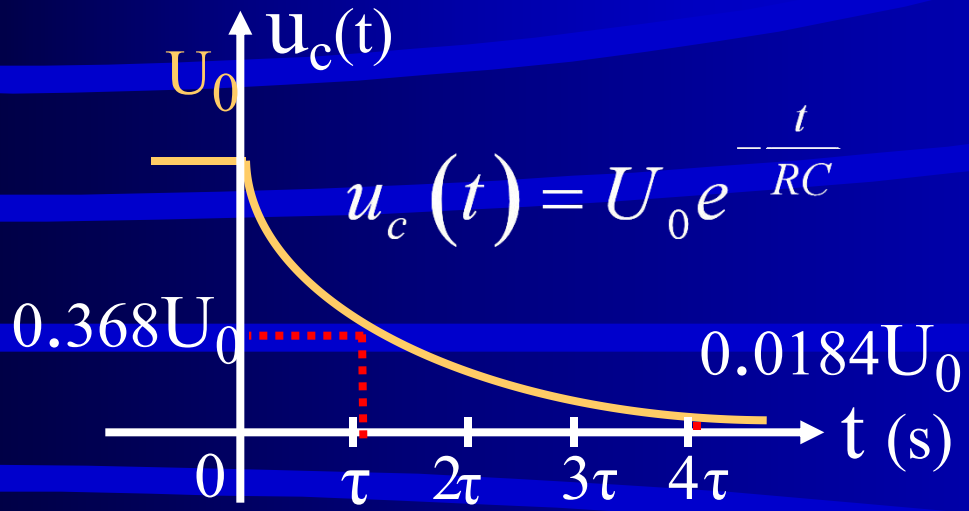
得：$\begin{cases} RC\dfrac{du_C}{dt} + u_C = 0 \qquad t \geq 0 \\ u_C(0) = U_0 \end{cases}$ 初始条件

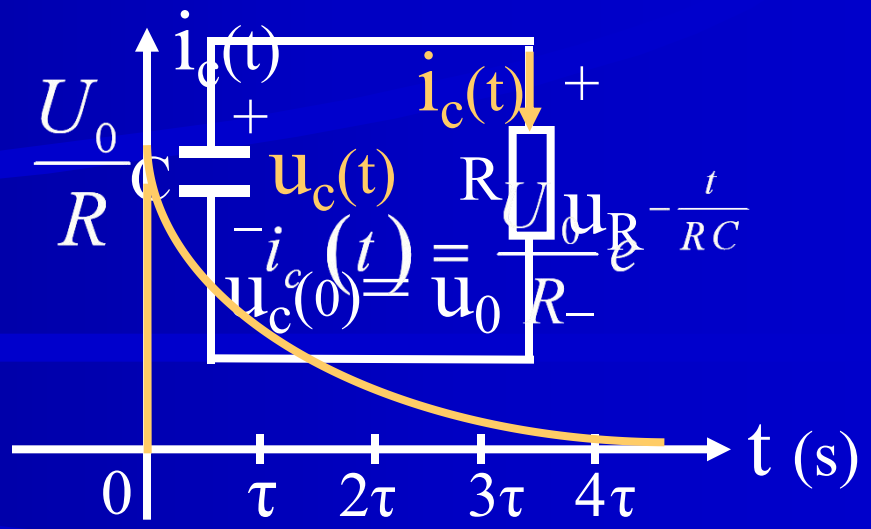
一阶齐次方程

特征方程：$RCP + 1 = 0 \qquad \therefore u_c(t) = Ae^{Pt} = Ae^{-\frac{t}{RC}} \qquad t \geq 0$

特征方程根：$P = -\dfrac{1}{RC} \qquad \because u_c(0) = Ae^{-\frac{1}{Rc}\times 0} = A = U_0$

∴解得：$u_c(t) = U_0 e^{-\frac{t}{RC}} \qquad t \geq 0$





$$i_c(t) = -C\frac{du_c(t)}{dt} = -C\frac{d}{dt}\left(U_0 e^{-\frac{t}{RC}}\right) = \frac{U_0}{R}e^{-\frac{t}{RC}} \qquad t \geq 0$$

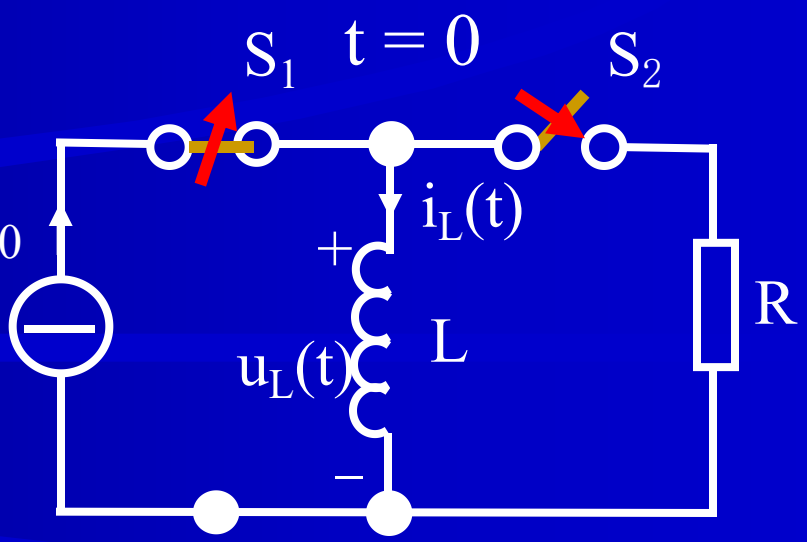
时间常数

u_c不能跃变，i_c可以跃变。

RC电路的零输入响应$u_c(t)$，$i_c(t)$都是按指数规律衰减的，衰减的快慢由 τ 决定。$\tau = RC$(秒)

τ 越小，$u_c(t)$，$i_c(t)$衰减的越快。

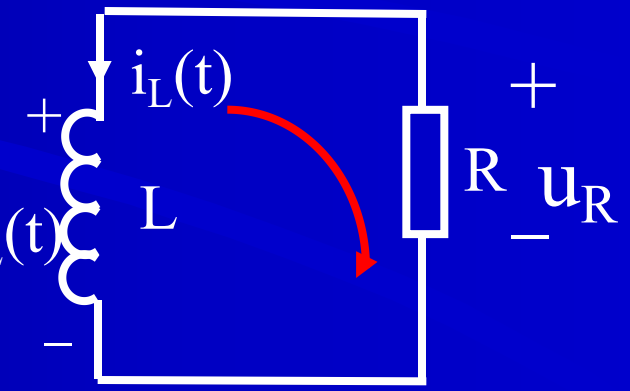
2. RL电路的零输入响应：



t <0时： $i_L(0)=I_0$ $I_s= I_0$

由KVL: $u_L-u_R= 0$ $t \geq 0$

由元件的性质：
$$\begin{cases} u_R = -R \times i_L \\ u_L = L\dfrac{di_L}{dt} \end{cases}$$
代入上式



得：
$$\begin{cases} L\dfrac{di_L}{dt} + Ri_L = 0 & t \geq 0 \\ i_L(0) = I_0 \end{cases}$$
初始条件

特征方程： $LP + R = 0$

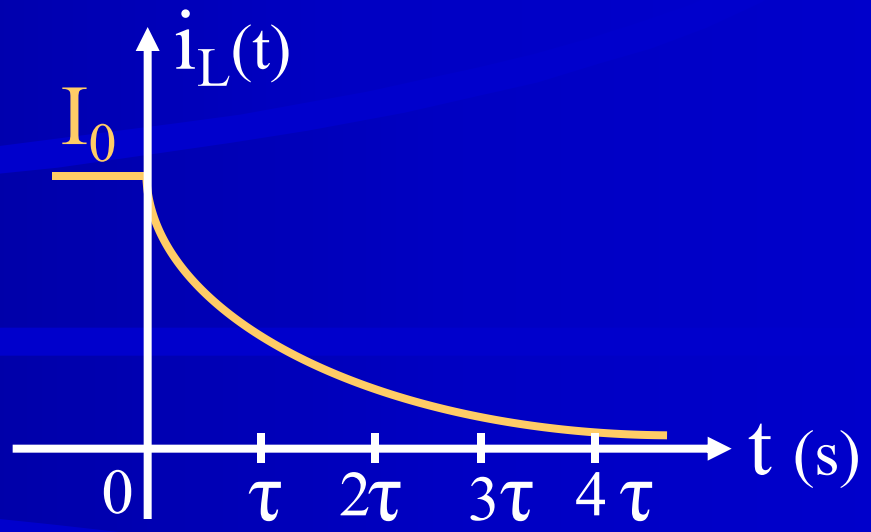
特征方程根： $P = -\dfrac{R}{L}$

$$\therefore i_L(t) = Ae^{Pt} = Ae^{-\frac{R}{L}t}$$

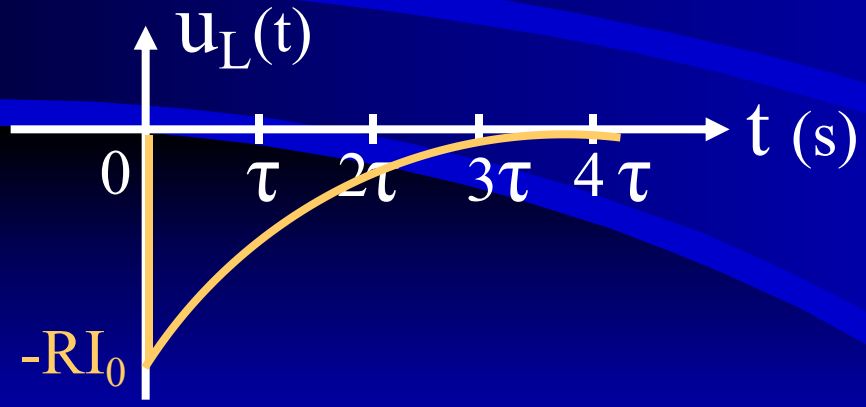
$$\because i_L(0) = Ae^{-\frac{R}{L}\times 0} = A = I_0$$

$\therefore$ 解得：

$$i_L(t) = I_0 e^{-\frac{R}{L}t} \qquad t \geq 0$$



$$u_L(t) = L\frac{di_L(t)}{dt} = L\frac{d}{dt}\left(I_0 e^{-\frac{R}{L}t}\right) = -RI_0 e^{-\frac{R}{L}t} \qquad t \geq 0$$



结论：i_L不能跃变，u_L可以跃变。

时间常数 $\tau = \frac{L}{R} = GL$（秒）

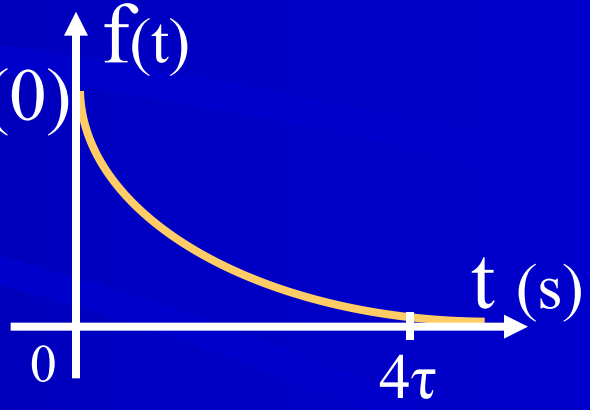
总结:

A. 零输入响应与$u_c(0)$, $i_L(0)$, τ 有关。

B. 零输入响应的一般形式: $f(t)=f(0)e^{-\frac{t}{\tau}} \quad t\geq 0$

$$u_c(t)=u_c(0)e^{-\frac{t}{RC}} \quad t\geq 0 \quad \tau=RC$$

$$i_L(t)=i_L(0)e^{-\frac{R}{L}t} \quad t\geq 0 \quad \tau=GL$$

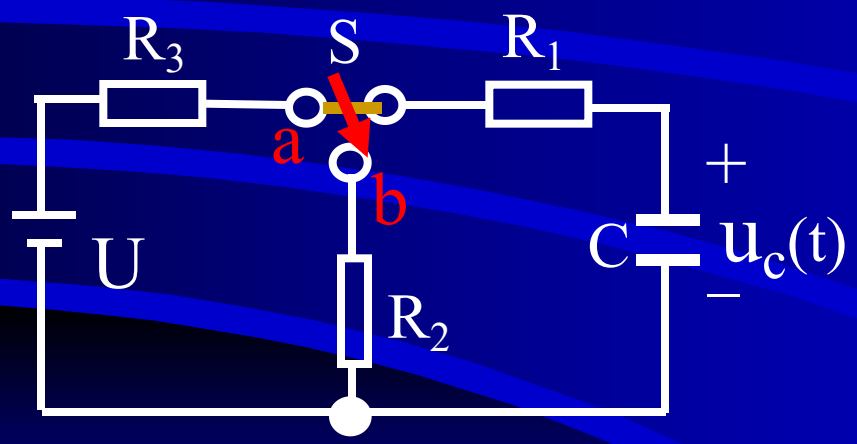


C. 零输入响应都是按指数规律衰减的，衰减的快慢由τ决定，τ越小，$u_c(t)$, $i_L(t)$衰减的越快。

D. 时间常数τ的求法:

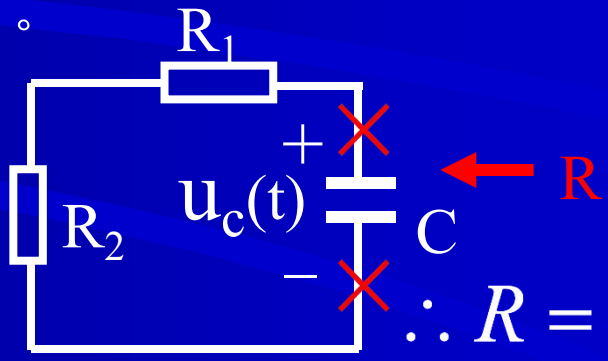
- 在换路后(即 $t \geq 0$)的电路中求。
- R是从动态元件两端看进去的戴维南等效电阻。

例4: $t \geq 0$时, S由a打向b点,求$u_c(t)$。



已知: $u_c(0) = U$

解: $t \geq 0$ 时的电路为:



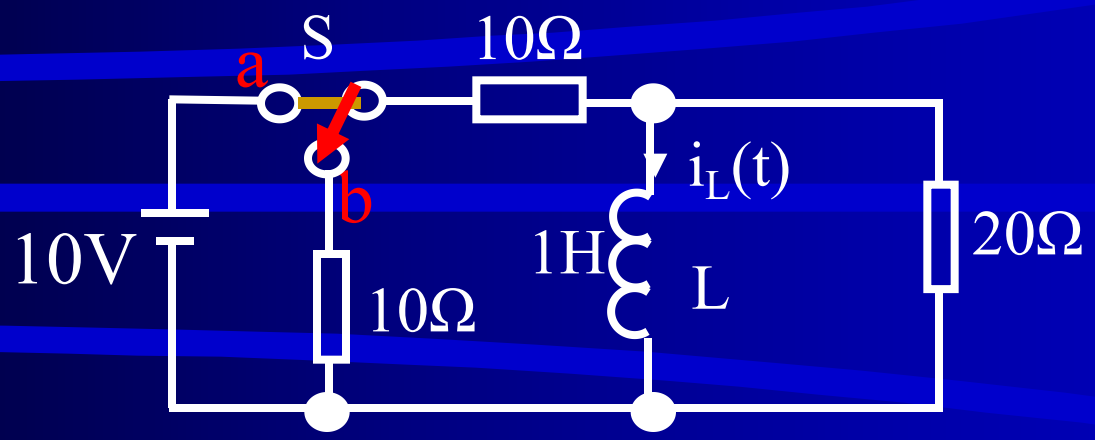
$$\therefore R = R_1 + R_2$$

$$\therefore \tau = (R_1 + R_2)C$$

$$\therefore u_c(t) = Ue^{-\frac{t}{(R_1+R_2)C}}$$

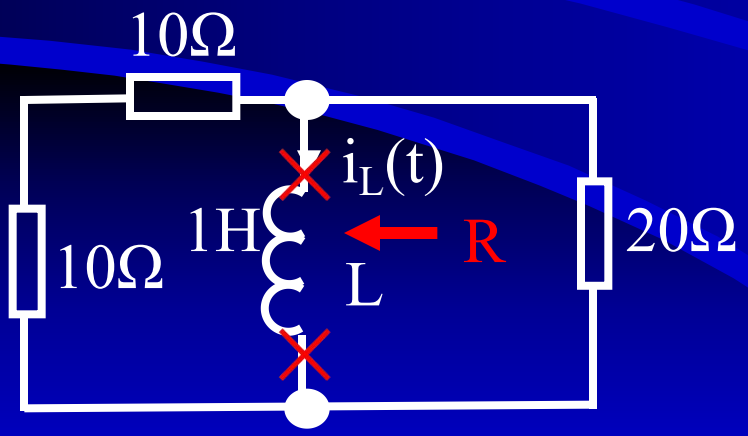
$$t \geq 0$$

例5: $t \geq 0$时, S由a打向b点,求$i_L(t)$。



已知: $i_L(0) = 1\text{A}$

解: $t \geq 0$ 时的电路为:



$$\therefore R = 20//(10+10) = 10\Omega$$

$$\therefore \tau = \frac{L}{R} = \frac{1}{10}\text{S}$$

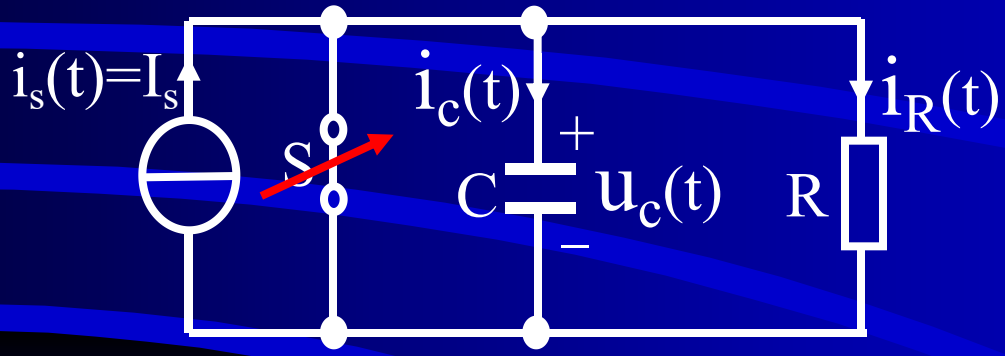
$$\therefore i_L(t) = e^{-10t}\text{A} \quad t \geq 0$$

5.3.2 零状态响应

动态元件本身没有储能$u_C(0)=0$,或$i_L(0)=0$,响应仅与外加激励有关。

1. RC电路零状态响应

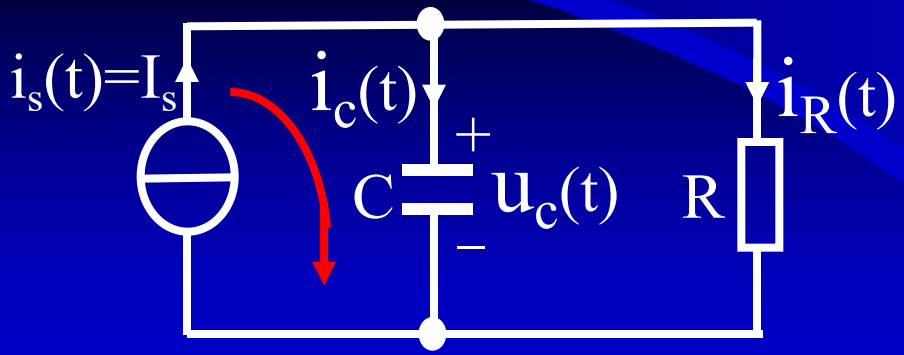
(1) RC(并联）电路的零状态响应



t = 0时, 开关打开.

因电容上的电压不能突变

$$\therefore u_c(0_+)=u_c(0_-)=0$$

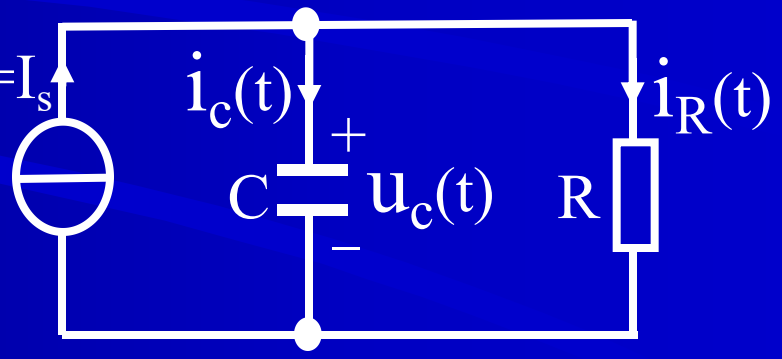


$\therefore t=0_+$时全部的电流都流过C,使C充电,R上没有电流。

$\because i_s(t) = I_s = C\dfrac{du_c}{dt}$　　电容电压的变化率：$\left.\dfrac{du_c}{dt}\right|_{0_+} = \dfrac{I_S}{C}$

$$u_C(t)\uparrow \rightarrow i_R(t) = \frac{u_C(t)}{R}\uparrow \rightarrow i_C(t)\downarrow \rightarrow i_C(\infty) = 0$$

∴t→∞时,C上没有电流，u_C恒定，C相当与开路，全部的电流都流过R。



由KCL：　$i_c + i_R = I_s$

由元件的性质：$\begin{cases} i_C = C\dfrac{du_c}{dt} \\ i_R = \dfrac{u_C}{R} \end{cases}$ 代入上式得

$$\begin{cases} C\dfrac{du_c}{dt} + \dfrac{u_c}{R} = I_s & t \geq 0 \\ u_c(0) = 0 \end{cases}$$

初始条件

一阶非齐次方程

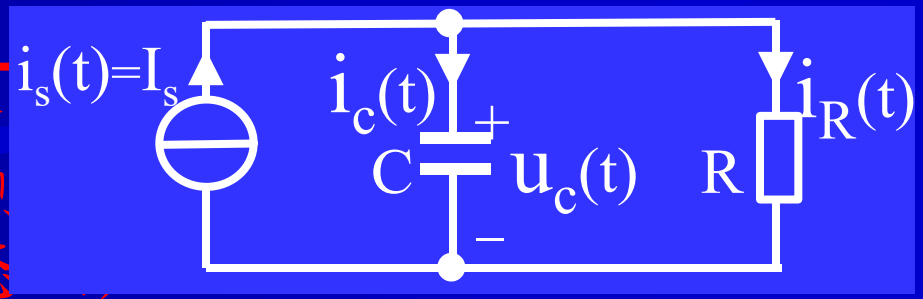
方程解有两部分：

$u_c(t) = u_{ch}(t) + u_{cp}(t)$

齐次方程解　非齐

固有响应　强制

暂态响应　稳态



齐次方程解：（暂态响应）$u_{ch}(t) = Ae^{-\frac{t}{\tau}}$

非齐次方程特解：（稳态响应）$u_{cp}(t) = u_c(\infty) = I_S R$

完全解：$u_c(t) = Ae^{-\frac{t}{\tau}} + I_S R$

稳态时，u_c恒定，电容相当于开路。

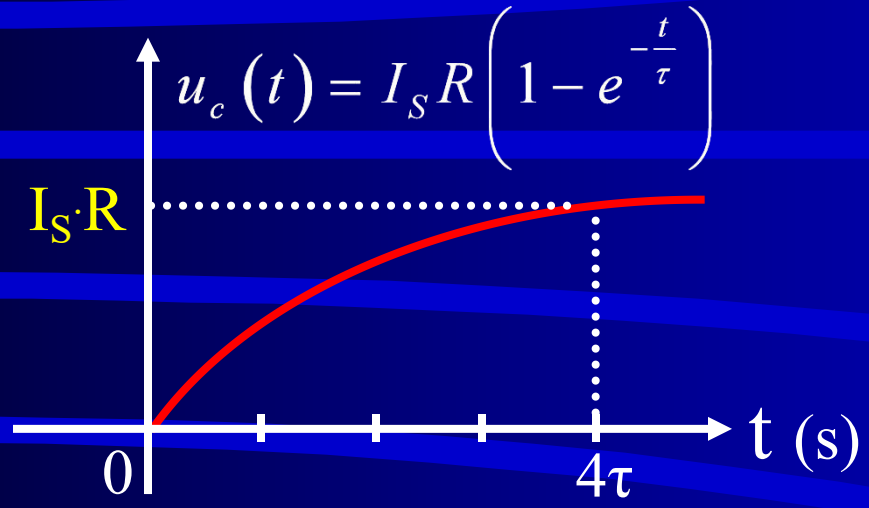
$\because u_c(0) = A + I_S R = 0$　解得：$A = -I_S R$

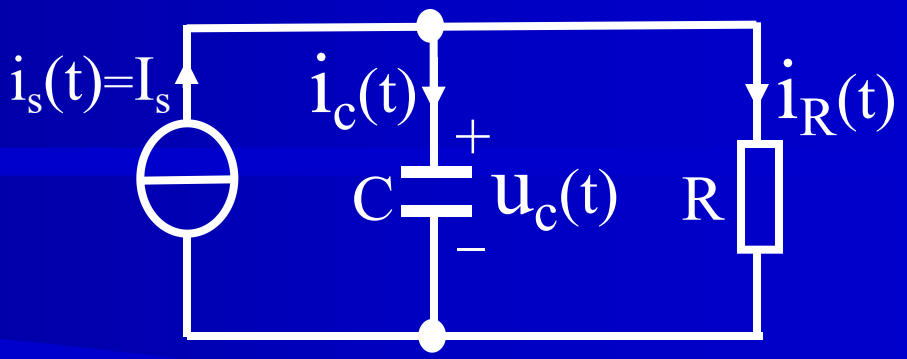
$$\therefore u_c(t) = -I_S R e^{-\frac{t}{\tau}} + I_S R = I_S R\left(1 - e^{-\frac{t}{\tau}}\right) = u_C(\infty)\left(1 - e^{-\frac{t}{\tau}}\right) \quad t \geq 0$$

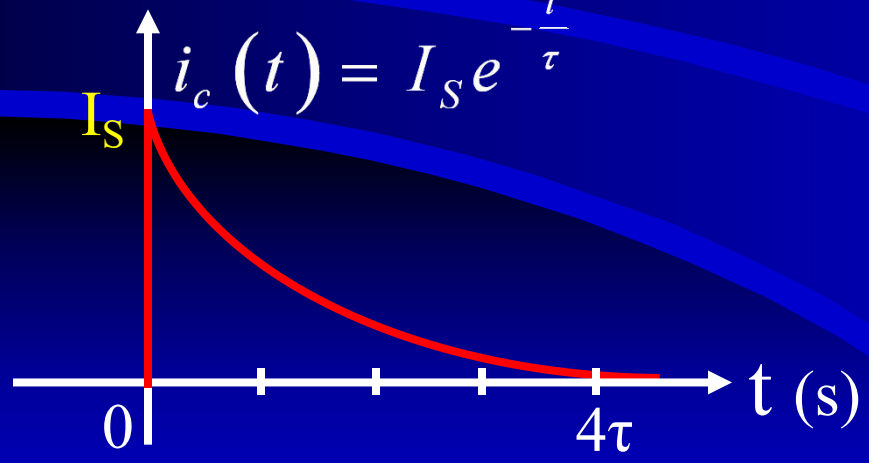
$$\therefore i_c(t) = C\frac{du_c}{dt} = C\frac{d}{dt}\left[I_S R\left(1 - e^{-\frac{t}{RC}}\right)\right] = I_S e^{-\frac{t}{RC}} \quad t \geq 0$$

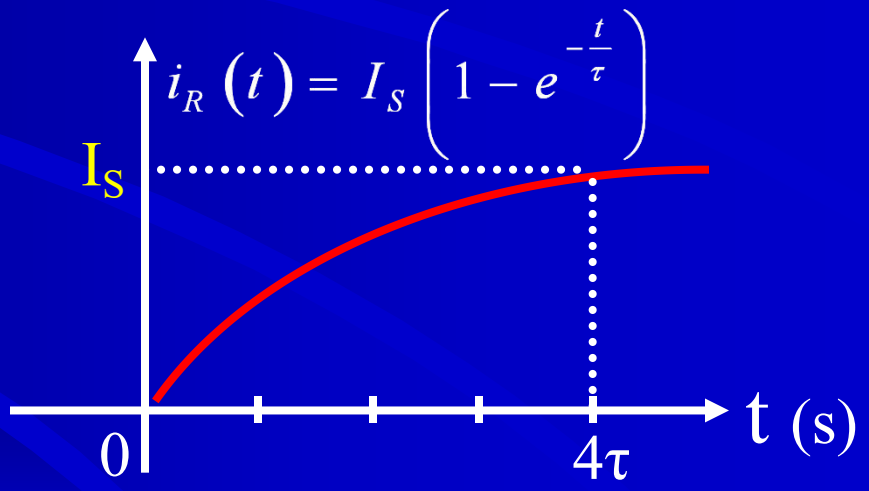
$$\therefore i_R(t) = \frac{u_c(t)}{R} = I_S\left(1 - e^{-\frac{t}{RC}}\right) \quad t \geq 0$$

电路中各输出波形：


$u_c(t) = I_S R\left(1 - e^{-\frac{t}{\tau}}\right)$
$I_S \cdot R$
0
4τ
t (s)

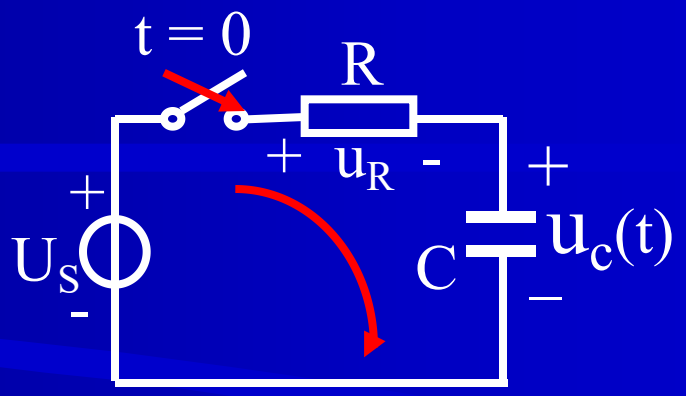

$i_s(t)=I_s$
$i_c(t)$
$i_R(t)$
+
C
$u_c(t)$
R
−


$i_c(t) = I_S e^{-\frac{t}{\tau}}$
I_S
0
4τ
t (s)


$i_R(t) = I_S\left(1 - e^{-\frac{t}{\tau}}\right)$
I_S
0
4τ
t (s)

(2) RC（串联）电路的零状态响应

由KVL: $u_R + u_c = U_S$



$$\therefore \begin{cases} RC\dfrac{du_c}{dt} + u_c = U_s & t \geq 0 \\ u_c(0) = 0 & \text{初始条件} \end{cases}$$

$$u_c(t) = u_{ch}(t) + u_{cp}(t)$$

暂态响应: $u_{ch}(t) = Ae^{-\frac{t}{\tau}}$

稳态响应: $u_{cp}(t) = u_c(\infty) = U_S$

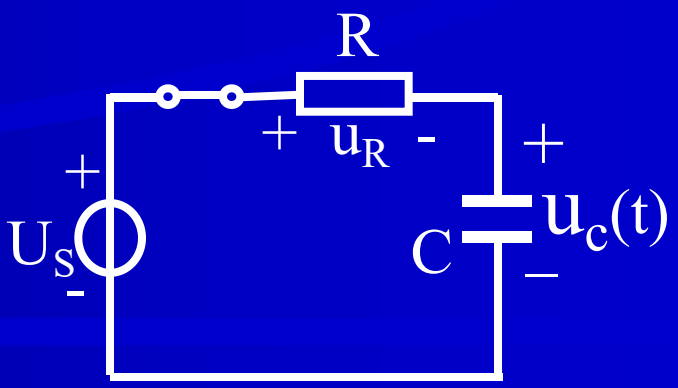
稳态时，u_c恒定，电容相当于开路。

完全解: $u_c(t) = Ae^{-\frac{t}{\tau}} + U_S$

$\because u_c(0) = A + U_S = 0$ 解得: $A = -U_S$

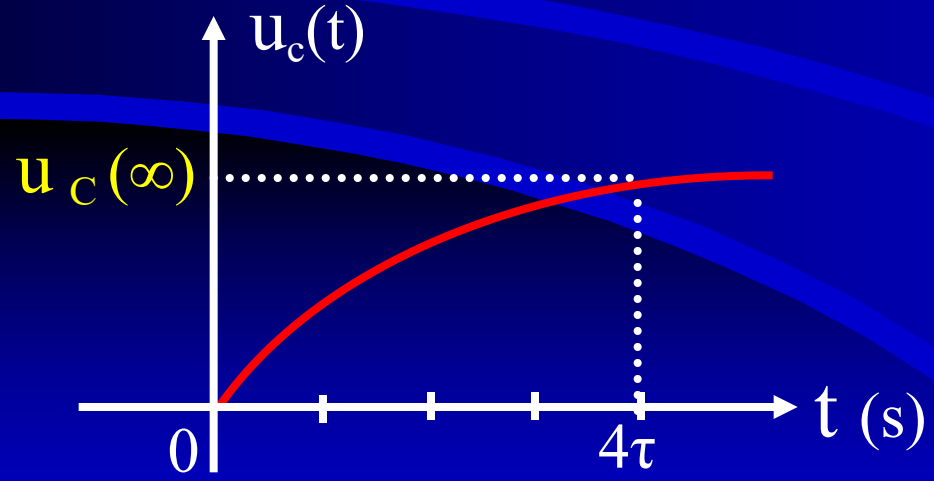
$$\therefore u_c(t) = -U_S e^{-\frac{t}{\tau}} + U_S = U_S\left(1 - e^{-\frac{t}{RC}}\right)$$

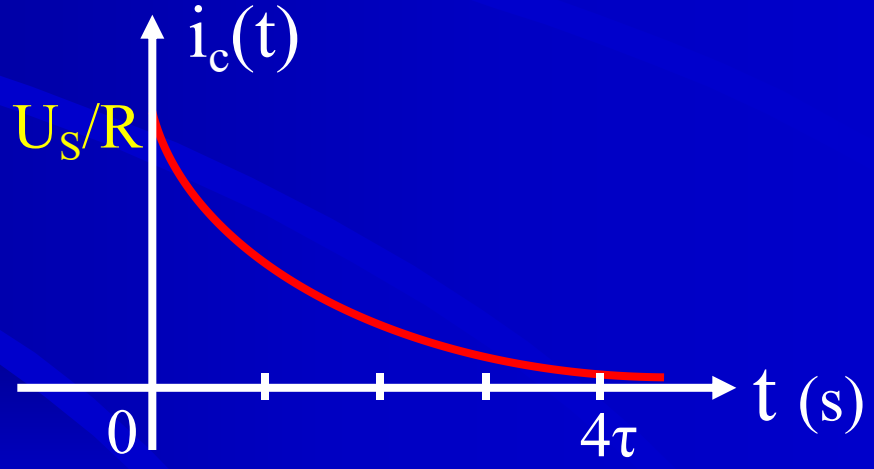
$$= u_C(\infty)\left(1 - e^{-\frac{t}{RC}}\right) \quad t \geq 0$$



$$\therefore i_c(t) = C\frac{du_c}{dt} = C\frac{d}{dt}\left[U_S\left(1 - e^{-\frac{t}{RC}}\right)\right] = \frac{U_S}{R}e^{-\frac{t}{RC}} \quad t \geq 0$$

$$\therefore i_R(t) = i_C(t) \quad t \geq 0$$

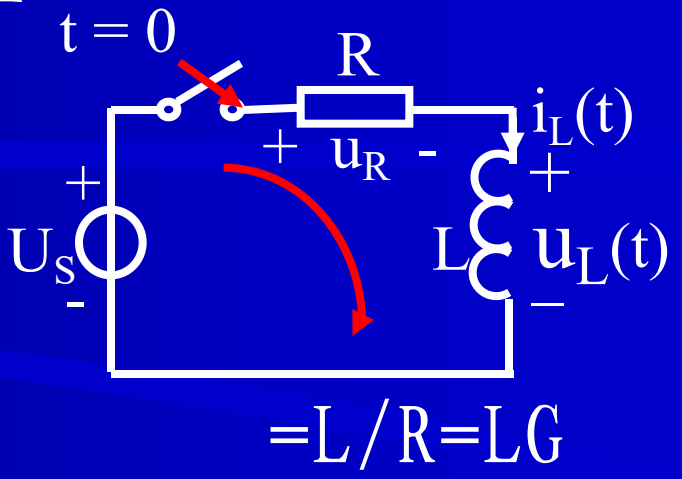




2. RL电路的零状态响应

(1) RL（串联）电路的零状态响应

由KVL: $u_{L+}\, u_R = U_S$



$$L\frac{di_L}{dt} + R\cdot i_L = U_s$$

$$\therefore \begin{cases} \dfrac{L}{R}\dfrac{di_L}{dt} + i_L = \dfrac{U_S}{R} & t \geq 0 \\ i_L(0) = 0 & \text{初始条件} \end{cases}$$

$$i_L(t) = i_{Lh}(t) + i_{Lp}(t)$$

暂态响应: $i_{Lh}(t) = Ae^{-\frac{t}{\tau}} = Ae^{-\frac{R}{L}t}$

稳态响应: $i_{Lp}(t) = i_L(\infty) = \dfrac{U_S}{R}$

完全解：$i_L(t) = Ae^{-\frac{t}{\tau}} + \frac{U_S}{R} \quad t \geq 0$

$\because i_L(0) = A + \frac{U_S}{R} = 0$ 解得：$A = -\frac{U_S}{R}$

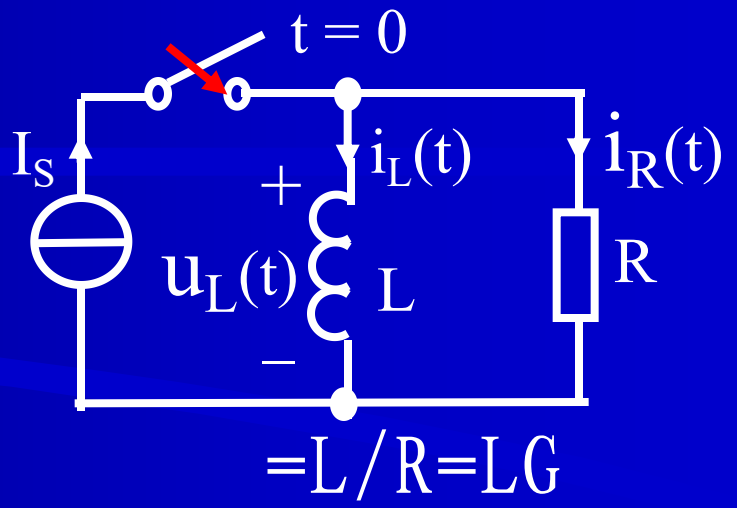
$$\therefore i_L(t) = -\frac{U_S}{R}e^{-\frac{t}{\tau}} + \frac{U_S}{R} = \frac{U_S}{R}\left(1 - e^{-\frac{t}{\tau}}\right)$$

$$= i_L(\infty)\left(1 - e^{-\frac{t}{\tau}}\right) \quad t \geq 0$$

$$\therefore u_L(t) = L\frac{di_L(t)}{dt} = L\frac{d}{dt}\left[\frac{U_S}{R}\left(1 - e^{-\frac{R}{L}t}\right)\right]$$

$$= U_S e^{-\frac{R}{L}t} \quad t \geq 0$$

(2) RL (并联) 电路的零状态响应



由KCL: $i_R + i_L = I_S$

$$\therefore \begin{cases} G \cdot L \dfrac{di_L}{dt} + i_L = I_S & t \geq 0 \\ i_L(0) = 0 & \text{初始条件} \end{cases}$$

解: $i_L(t) = i_{Lh}(t) + i_{Lp}(t)$

暂态响应: $i_{Lh}(t) = Ae^{-\frac{t}{\tau}} = Ae^{-\frac{R}{L}t}$

稳态响应: $i_{Lp}(t) = i_L(\infty) = I_S$

完全解：$i_L(t) = Ae^{-\frac{t}{\tau}} + I_S \quad t \geq 0$

$\because i_L(0) = A + I_S = 0$ 解得：$A = -I_S$

$$\therefore i_L(t) = -I_S e^{-\frac{t}{\tau}} + I_S = I_S\left(1 - e^{-\frac{t}{\tau}}\right)$$

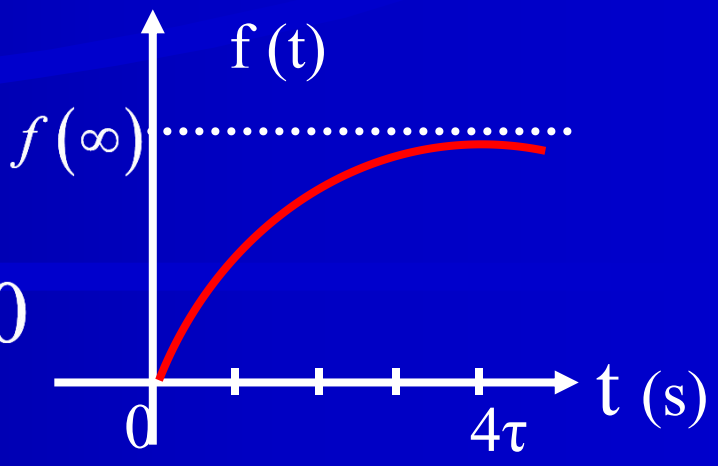
$$= i_L(\infty)\left(1 - e^{-\frac{R}{L}t}\right) \quad t \geq 0$$

$$\therefore u_L(t) = L\frac{di_L(t)}{dt} = L\frac{d}{dt}\left[I_S\left(1 - e^{-\frac{R}{L}t}\right)\right]$$

$$= L \cdot I_S\left(-e^{-\frac{R}{L}t} \cdot -\frac{R}{L}\right) = R \cdot I_S e^{-\frac{R}{L}t} \quad t \geq 0$$

总结:

A.零状态响应的一般形式:



$$f(t)=f(\infty)\left(1-e^{-\frac{t}{\tau}}\right) \qquad t\geq 0$$

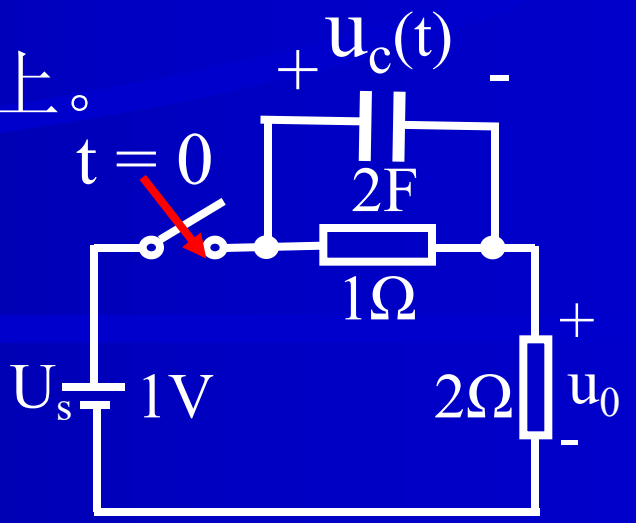
$$\therefore u_c(t)=u_c(\infty)\left(1-e^{-\frac{t}{RC}}\right) \qquad t\geq 0 \qquad \tau=RC$$

$$\therefore i_L(t)=i_L(\infty)\left(1-e^{-\frac{R}{L}t}\right) \qquad t\geq 0 \qquad \tau=GL$$

B.零状态响应都是按指数规律上升的，上升的快慢由 τ 决定，τ 越小，$u_c(t)$，$i_L(t)$ 上升的越快。

例6: 已知$u_C(0)=0$,$t=0$时开关合上。

求：$t>0$ 时，$u_C(t)$和$u_0(t)$。



解法一:

$$\because u_c(t)=u_c(\infty)\left(1-e^{-\frac{t}{\tau}}\right) \qquad t\geq 0$$

$$\tau=RC=(2//1)\cdot 2=\frac{4}{3}\text{S} \qquad u_c(\infty)=\frac{1}{3}\text{V}$$

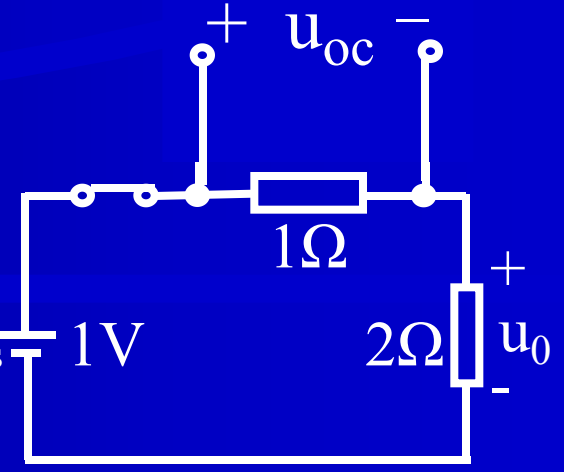
$$\therefore u_c(t)=\frac{1}{3}\left(1-e^{-\frac{3}{4}t}\right) \qquad t\geq 0$$

$$u_0(t)=U_S-u_C(t)=1-\frac{1}{3}(1-e^{-\frac{3}{4}t})=(\frac{2}{3}+\frac{1}{3}e^{-\frac{3}{4}t})\,\text{V}$$

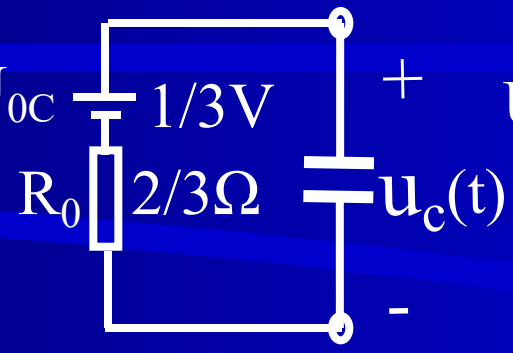
$$t\geq 0$$

解法二:

$t \geq 0$ 时的戴维南等效电路:



$$U_{0C} = \frac{1}{1+2} \cdot 1 = \frac{1}{3}\text{V}$$



$$R_0 = 1//2 = \frac{2}{3}\Omega$$

$$\therefore \tau = RC = \frac{2}{3} \cdot 2 = \frac{4}{3}\text{S} \qquad u_c(\infty) = \frac{1}{3}\text{V}$$

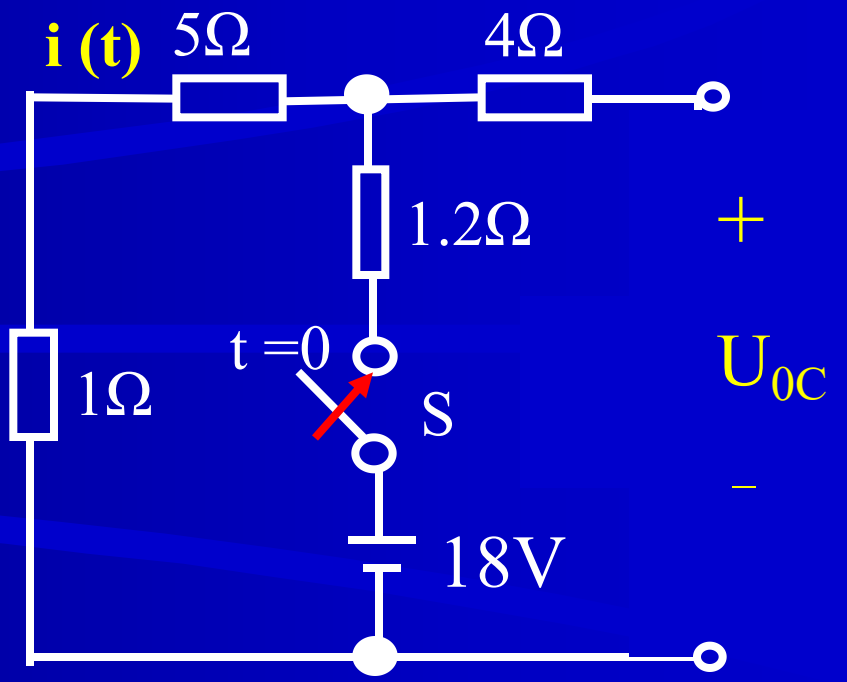
$$\therefore u_c(t) = \frac{1}{3}\left(1 - e^{-\frac{3}{4}t}\right) \qquad t \geq 0$$

求$u_0(t)$要回到原电路

$$u_0(t) = U_S - u_C(t) = \left(\frac{2}{3} + \frac{1}{3}e^{-\frac{3}{4}t}\right)\text{V} \qquad t \geq 0$$

例7: 已知 t = 0时开关合上。

求：t > 0 时的$i_L(t)$, i(t)。



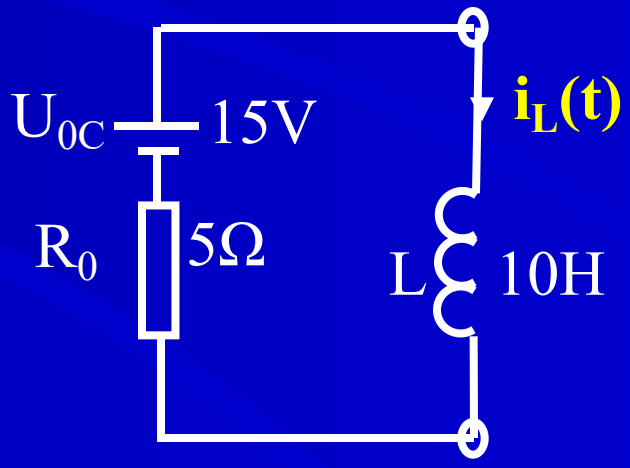
解: t > 0 时,从L两端看进去的戴维南等效电路：

$$U_{0C}=\frac{6}{1+5+1.2}\cdot 18=15\text{V}$$

$$R_0=4+1.2//(1+5)=5\Omega$$

$$\tau=\frac{L}{R}=\frac{10}{5}=2\text{S}\qquad i_L(\infty)=\frac{15}{5}=3\text{A}$$

$$\therefore i_L(t)=3\left(1-e^{-\frac{1}{2}t}\right)\text{A}\qquad t\geq 0$$



求i(t)要回到原电路

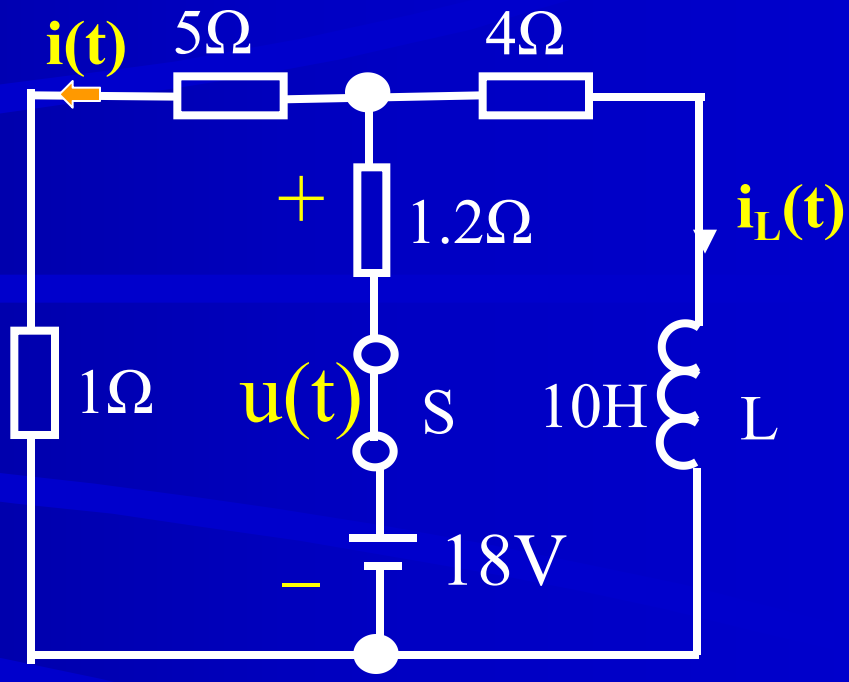
$$u(t)=4i_L(t)+10\frac{di_L(t)}{dt}$$

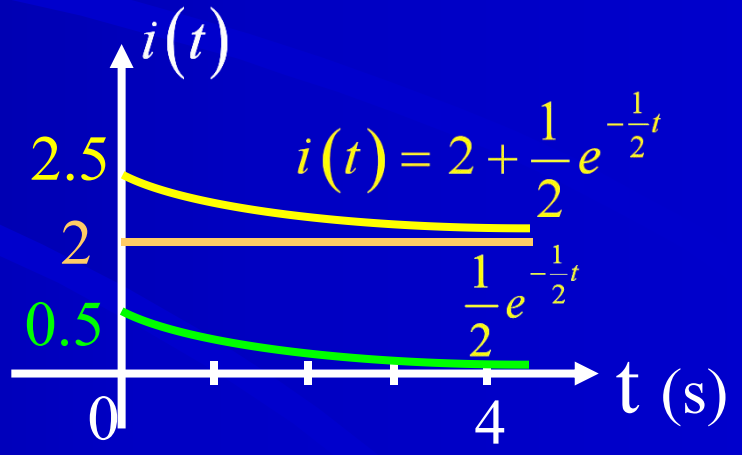
$$=4\cdot 3\left(1-e^{-\frac{1}{2}t}\right)+10\left(3\cdot -e^{-\frac{1}{2}t}\cdot -\frac{1}{2}\right)$$

$$=12\left(1-e^{-\frac{1}{2}t}\right)+15e^{-\frac{1}{2}t}$$

$$=12+3e^{-\frac{1}{2}t}\qquad t\geq 0$$

$$\therefore i(t)=\frac{u(t)}{1+5}=2+\frac{1}{2}e^{-\frac{1}{2}t}\qquad t\geq 0$$





5.3.3 全响应和三要素法

由动态元件的初始储能和外加激励共同作用产生的响应称为全响应。

全响应的一般形式为

$$f(t)=f(0_+)e^{-\frac{t}{\tau}}+f(\infty)\left(1-e^{-\frac{t}{\tau}}\right)$$

三要素为：

初始值 $f(0_+)$

$$u_c(0_+)=u_c(0_-)$$

$$i_L(0_+)=i_L(0_-)$$

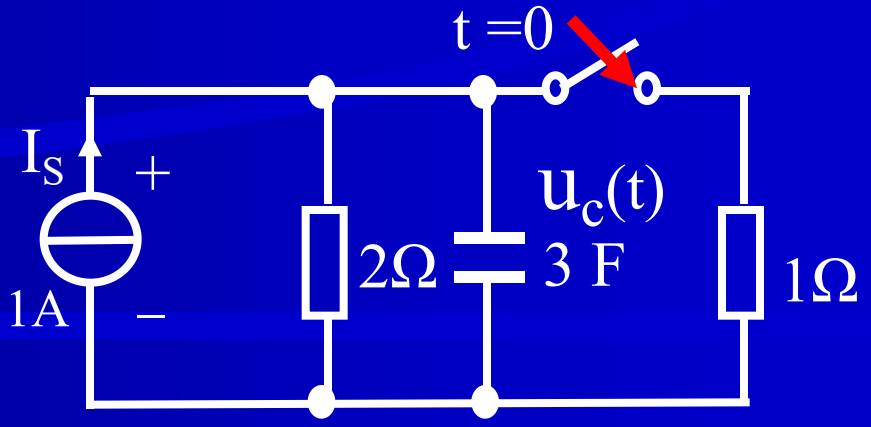
稳态值 $f(\infty)$

稳态时，电容相当于开路。

稳态时，电感相当于短路。

时间常数 τ　　$\tau=RC$　$\tau=GL$　（同一电路 τ 相同）

例8： 已知开关闭合前电路已达到稳定，用三要素法求$u_C(t)$。



解： 1. 求$u_C(0_+)$：

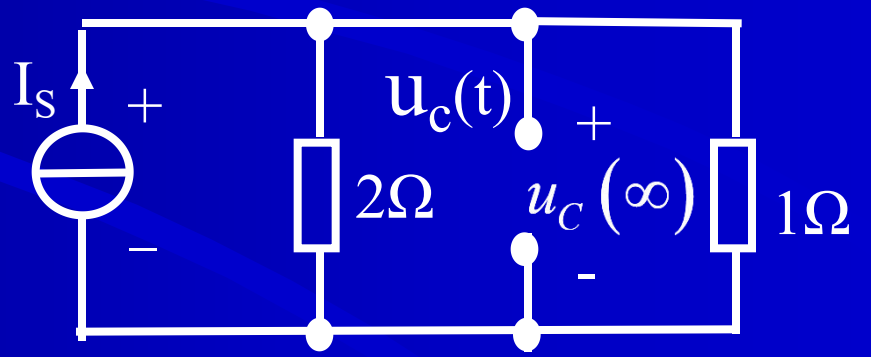
t =0_时，电路已达到稳定，即t→∞ 时，C→开路。

$$\therefore u_C(0_-) = 1 \cdot 2 = 2\text{V}$$

$$\therefore u_C(0_+) = u_C(0_-) = 2\text{V}$$

t→∞ 时，C→开路

2. 求 $u_C(\infty)$：

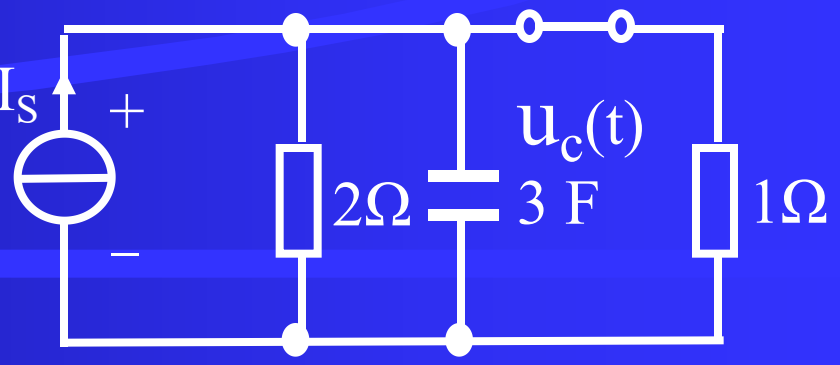


$$u_C(\infty) = \frac{2 \cdot 1}{2+1} \cdot 1 = \frac{2}{3}\text{V}$$

3. 求 τ：

$$R_0 = 2//1 = \frac{2}{3}\Omega$$

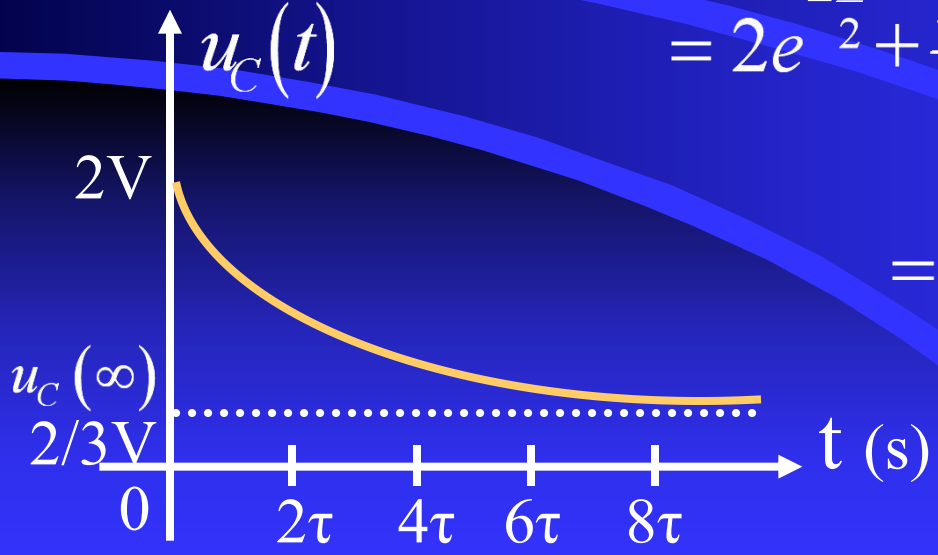
$$\therefore \tau = RC = \frac{2}{3}\cdot 3 = 2\mathrm{S}$$



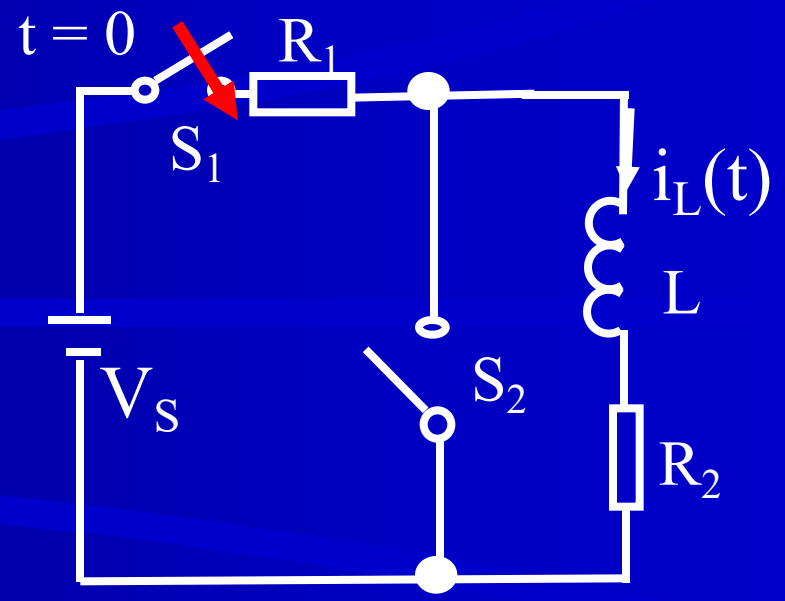
$$u_C(t) = u_C(0_+)e^{-\frac{t}{\tau}} + u_C(\infty)\left(1 - e^{-\frac{t}{\tau}}\right)$$

$$= 2e^{-\frac{t}{2}} + \frac{2}{3}(1 - e^{-\frac{t}{2}})$$

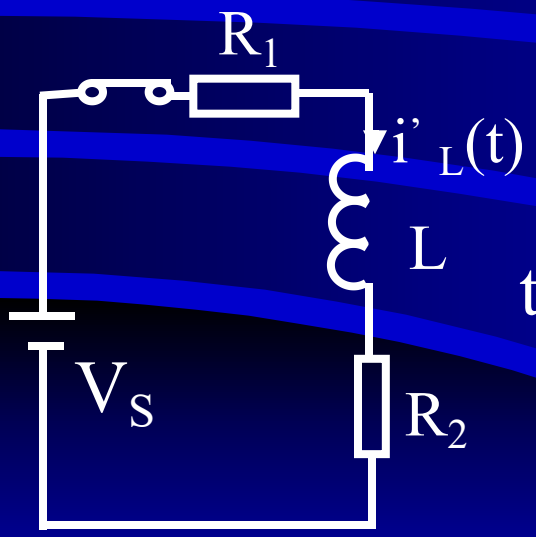
$$= (\frac{2}{3} + \frac{4}{3}e^{-\frac{t}{2}})\mathrm{V} \quad t \geq 0$$



例9： 已知：$t=0$时S_1合上，$t=t_1$时S_2合上。求$i_L(t)$，并画出波形。



解： 1. $t=0$时S_1合上



$t=0_-$ 时：$i_L(0_-)=0 \therefore i_L(0_+)=i_L(0_-)=0$

$t\to\infty$ 时，L→短路 $\therefore i_L(\infty)=\dfrac{V_S}{R_1+R_2}$

$$\therefore i_L'(t)=\frac{V_S}{R_1+R_2}\left(1-e^{-\frac{R_1+R_2}{L}t}\right) \quad 0\le t\le t_1$$

$$\tau_1=\frac{L}{R_1+R_2}$$

2. $t = t_1$时S_2合上

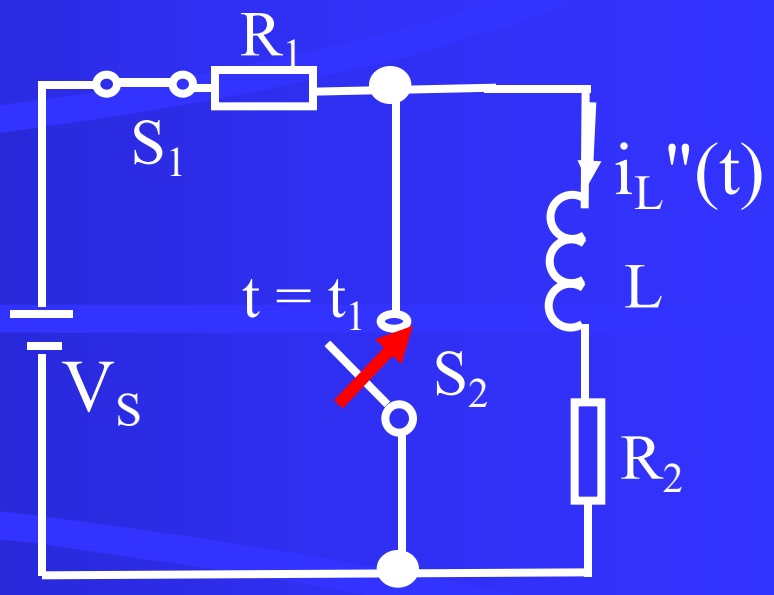
设S_2合上前为：$t = 0_-$

$$i_L(0_-) = \frac{V_S}{R_1 + R_2}\left(1 - e^{-\frac{R_1+R_2}{L}t_1}\right)$$

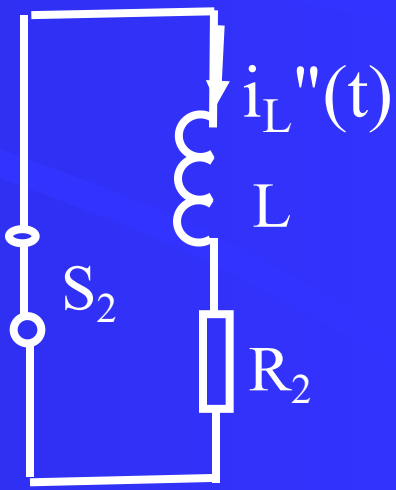
$$\therefore i_L(0_+) = \frac{V_S}{R_1 + R_2}\left(1 - e^{-\frac{R_1+R_2}{L}t_1}\right)$$

$t \to \infty$ 时，L→短路　　$\therefore i_L(\infty) = 0$

$$\therefore i_L''(t) = \frac{V_S}{R_1 + R_2}\left(1 - e^{-\frac{R_1+R_2}{L}t_1}\right) e^{-\frac{R_2}{L}(t-t_1)}$$



$$\tau_2 = \frac{L}{R_2}$$



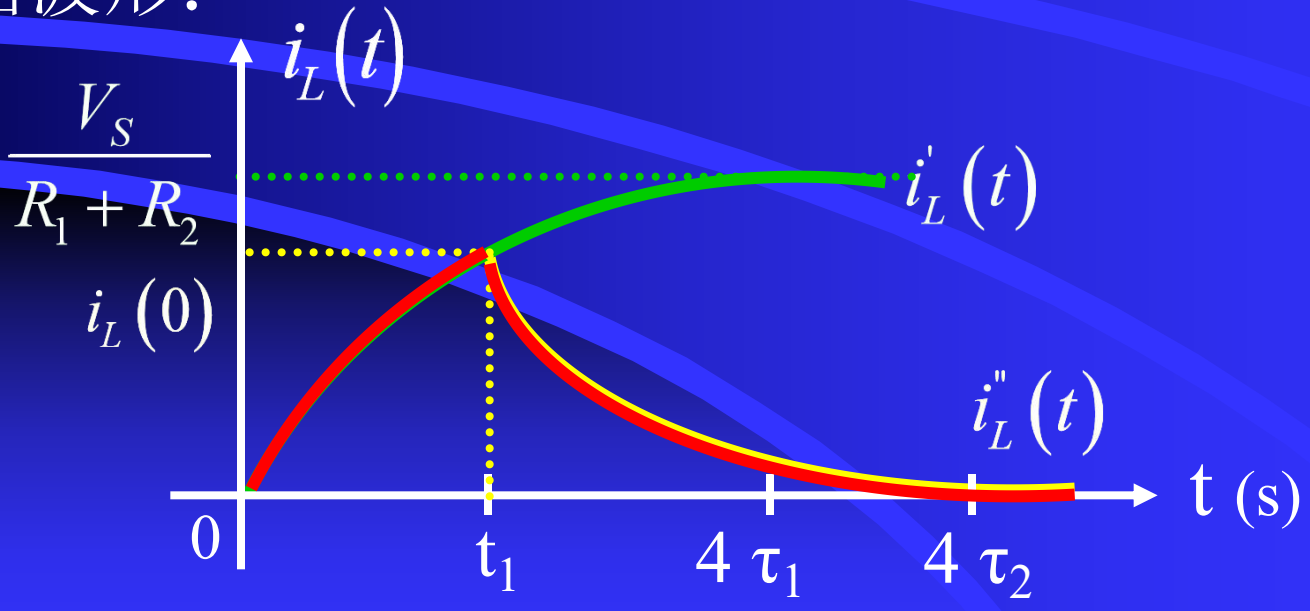
$t \geq t_1$

$$i_L'(t)=\frac{V_S}{R_1+R_2}\left(1-e^{-\frac{R_1+R_2}{L}t}\right) \qquad 0\le t\le t_1$$

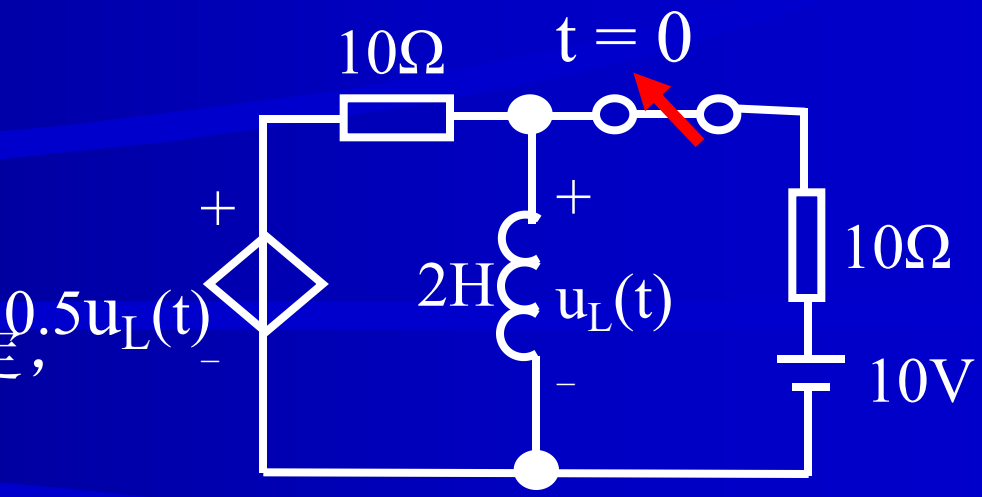
$$i_L''(t)=\frac{V_S}{R_1+R_2}\left(1-e^{-\frac{R_1+R_2}{L}t_1}\right)e^{-\frac{R_2}{L}(t-t_1)} \qquad t\ge t_1$$

$i_L(0)$

输出波形：



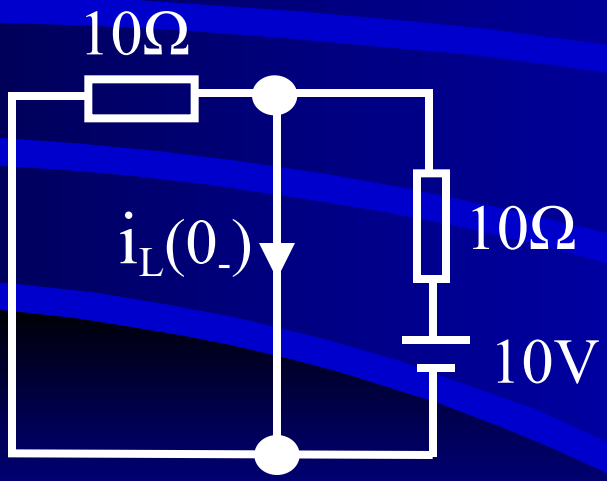
例10：　求 $t>0$ 时，$u_L(t)$。



解：1.　先求$u_L(0_+)$

(a). $t=0_-$ 时，电路已达到稳定，即$t\to\infty$ 时，L→短路。

$u_L(0_-)=0$，∴受控源也为零。

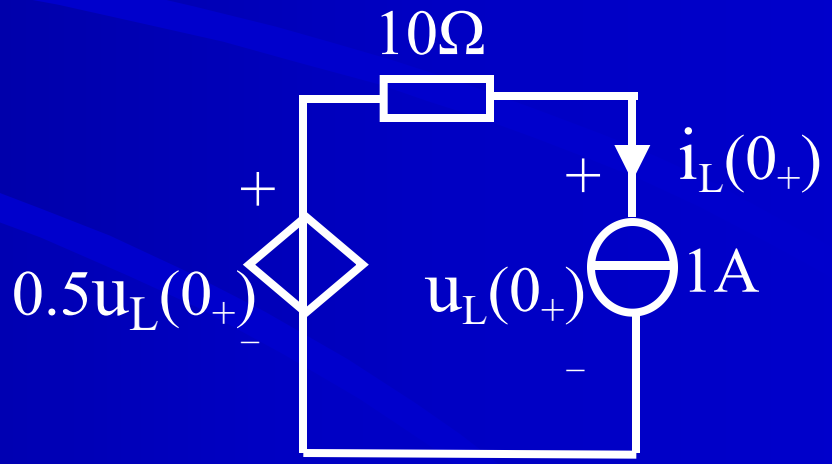


$$i_L(0_-)=\frac{10}{10}=1\text{A}$$

$$\therefore i_L(0_+)=i_L(0_-)=1\text{A}$$

(b). 作$t=0_+$ 时刻的等效电路。

（L→1A的电流源）



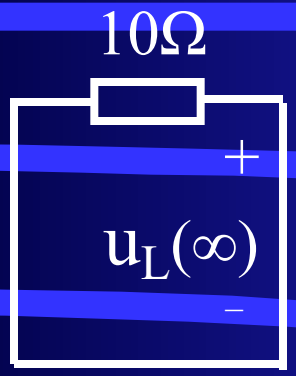
$$u_L(0_+)=-i_L(0_+)\cdot 10+0.5u_L(0_+)$$

$$\therefore u_L(0_+)=-20\text{V}$$

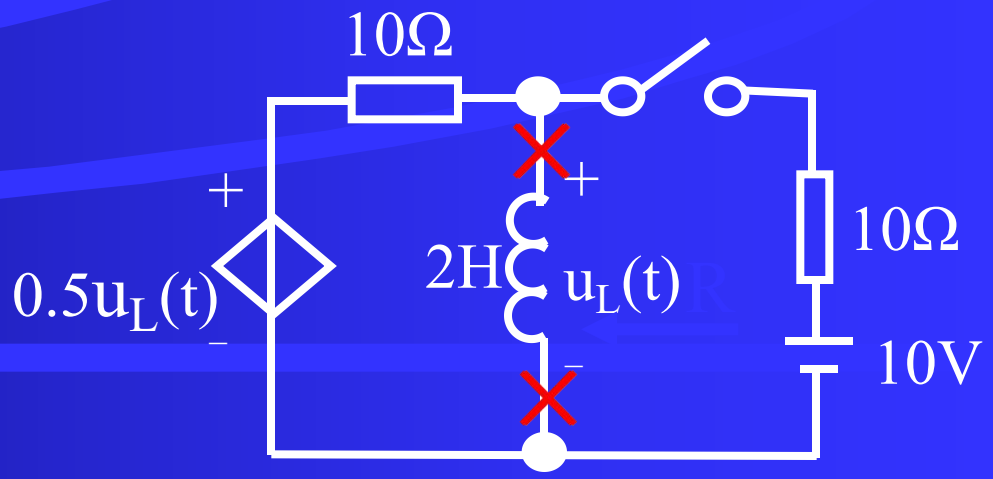
2. 求$u_L(\infty)$

t→∞ 时，L→短路。

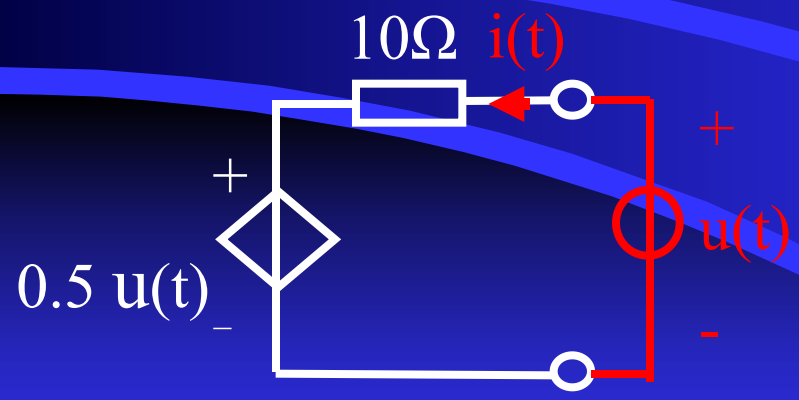
∴受控源也为零。



$$\therefore u_L(\infty)=0$$



3. 求τ　(a). 先求R

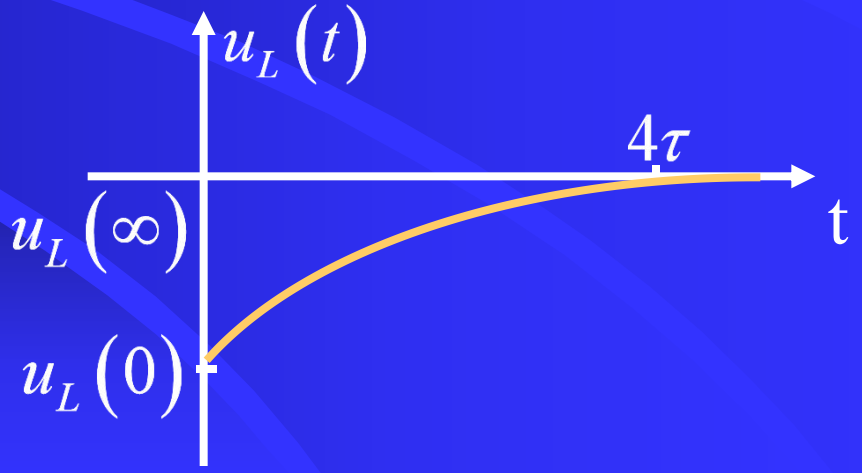


$$R=\frac{u(t)}{i(t)}$$

$$u(t)=i(t)\cdot 10+0.5u(t)$$

$$\therefore R=\frac{u(t)}{i(t)}=20\Omega \qquad \therefore \tau=\frac{L}{R}=\frac{2}{20}=\frac{1}{10}\text{S}$$

$$\therefore u_L(t)=u_L(0)e^{-\frac{t}{\tau}}=-20e^{-10t}\text{V}\quad t\geq 0$$



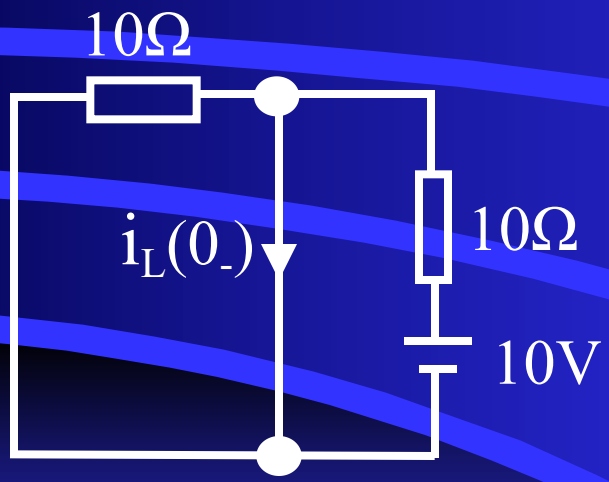
另解：先求$i_L(t)$,再求$u_L(t)$。

1. 求$i_L(0_+)$

$t=0_-$ 时，电路达稳定，

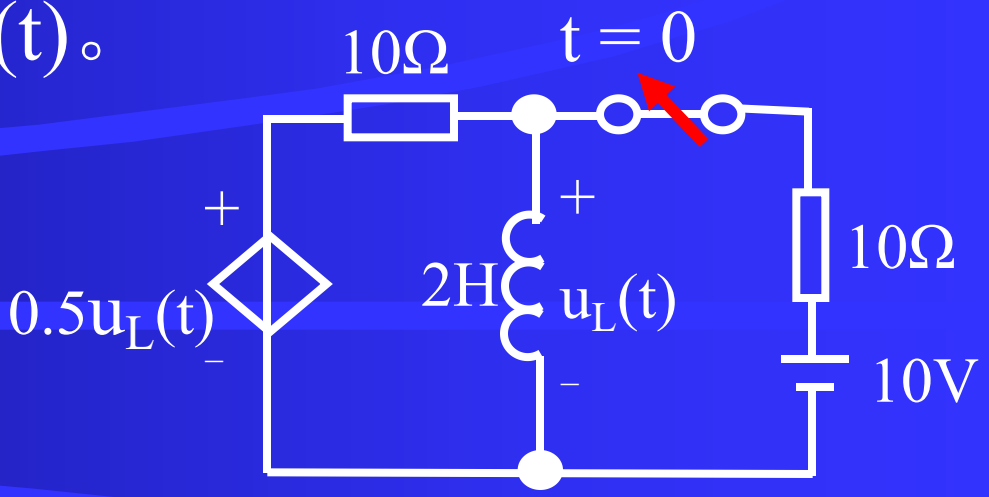
即$t\to\infty$ 时，L→短路。

$u_L(0_-)=0$, ∴受控源也为零。



$$i_L(0_-)=\frac{10}{10}=1\text{A}$$

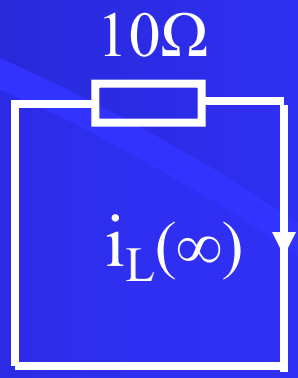
$$\therefore i_L(0_+)=i_L(0_-)=1\text{A}$$



2. 求$i_L(\infty)$

$t\to\infty$ 时，L→短路。

∴受控源也为零。



$$\therefore i_L(\infty)=0$$

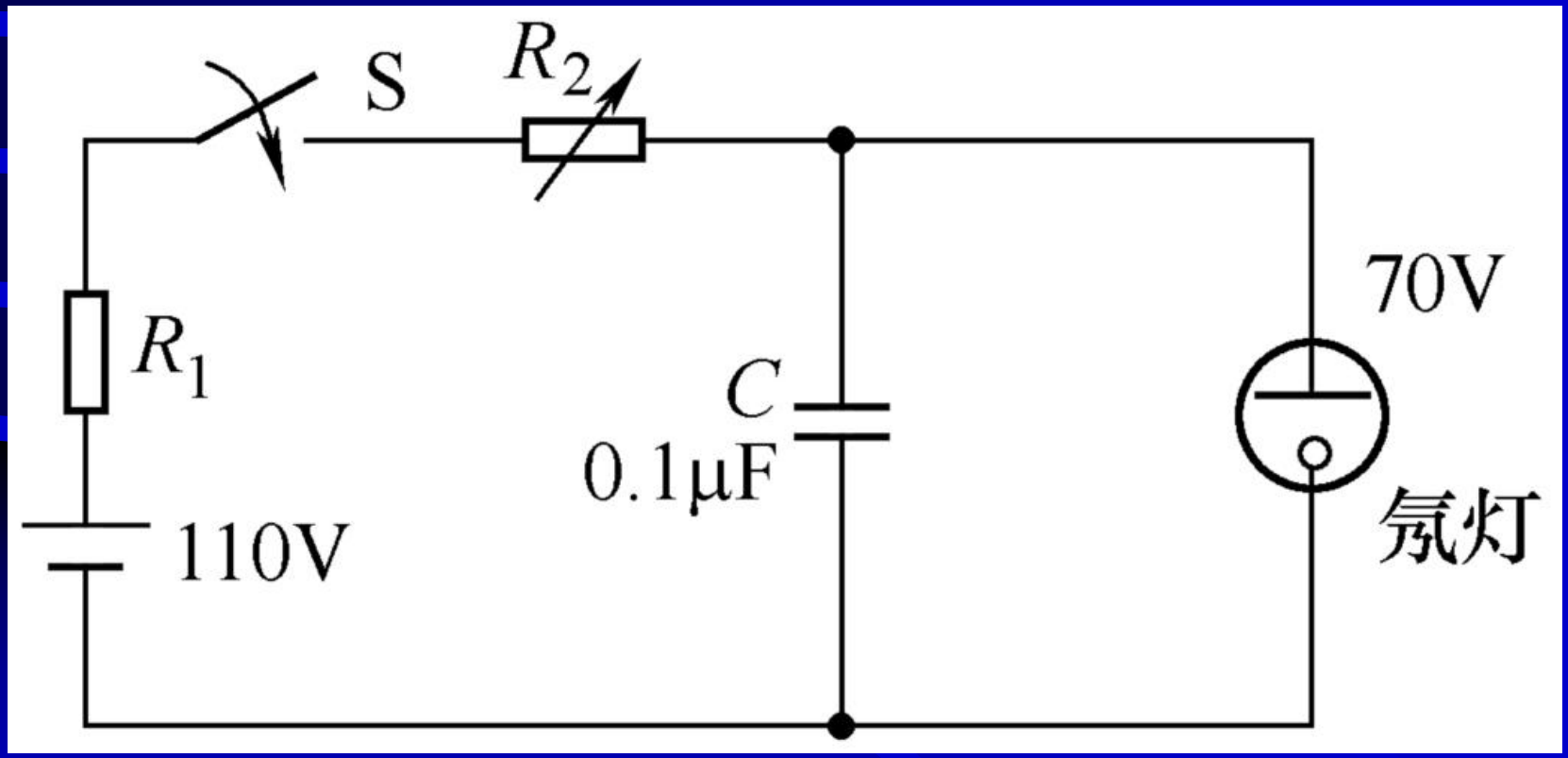
3. 求τ　　方法同上：　　$\therefore \tau = \frac{L}{R} = \frac{1}{10}\mathrm{S}$

$$\therefore i_L(t) = i_L(0)e^{-\frac{t}{\tau}} = e^{-10t}\mathrm{A} \quad t \geq 0$$

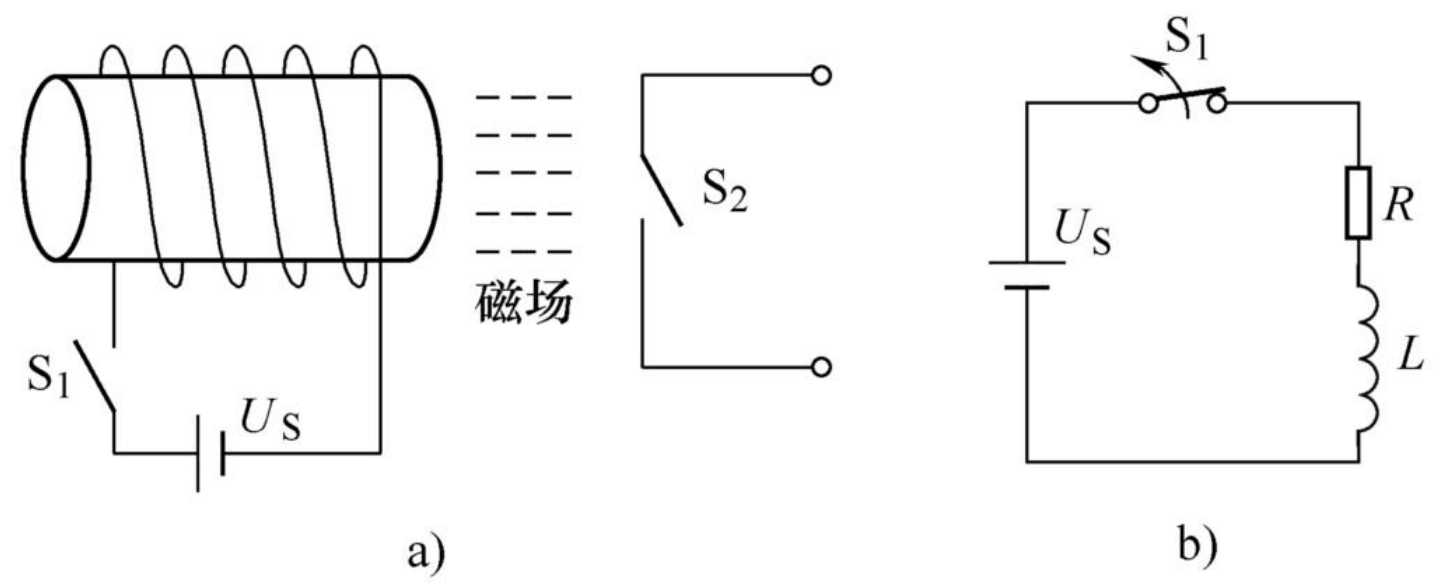
$$u_L(t) = L\frac{di_L(t)}{dt} = -20e^{-10t}\mathrm{V} \quad t \geq 0$$

5.3.4 一阶电路的实际应用

1. 闪烁灯电路



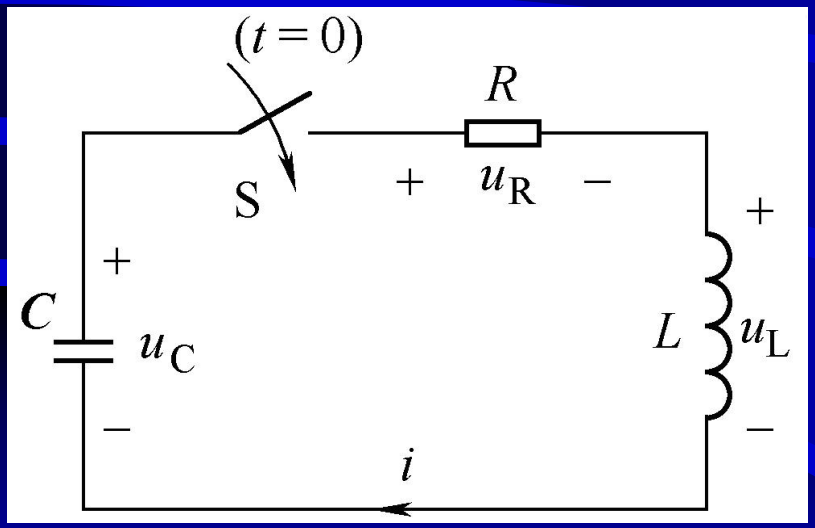
2. 继电器电路



5.4 二阶电路的分析

二阶电路: 电路中有二个动态元件，且电路方程用二阶微分方程来描述。

5.4.1 二阶电路的零输入响应



(串联电路选u_C为求解变量)

由KVL: $u_L + u_R + u_C = 0$

代入上式

$$\begin{cases} u_R = R \cdot i = R \cdot C \dfrac{du_C}{dt} \\ u_L = L \dfrac{di}{dt} = L \cdot C \dfrac{d^2 u_C}{dt^2} \end{cases}$$

$$\therefore LC \frac{d^2 u_C}{dt^2} + RC \frac{du_C}{dt} + u_C = 0$$

特征方程为：$P^2+\frac{R}{L}P+\frac{1}{LC}=0$

特征方程根为：$P_{1,2}=-\frac{R}{2L}\pm\sqrt{\left(\frac{R}{2L}\right)^2-\frac{1}{LC}}$

电路中R、L、C的参数不同，固有频率P_1, P_2可能有3种不同的情况。

$\left(\frac{R}{2L}\right)^2>\frac{1}{LC}$时，即$R>2\sqrt{\frac{L}{C}}$　P_1、P_2为不相同的负实数；（过阻尼）

$\left(\frac{R}{2L}\right)^2=\frac{1}{LC}$时，即$R=2\sqrt{\frac{L}{C}}$　P_1、P_2为相同的负实数；（临界阻尼）

$\left(\frac{R}{2L}\right)^2<\frac{1}{LC}$时，即$R<2\sqrt{\frac{L}{C}}$　P_1、P_2为共轭复数；（欠阻尼）

1. 过阻尼情况 当$R>2\sqrt{\frac{L}{C}}$时： P_1、P_2是不相同的负实数；

齐次方程解： $u_C(t)=A_1e^{P_1t}+A_2e^{P_2t}$

A_1, A_2由初始条件定：

$$\begin{cases} u_C(0)=A_1+A_2 \\ u_C'(0)=A_1P_1+A_2P_2=\dfrac{i_L(0)}{C} \end{cases}$$

$$\therefore A_1=\frac{1}{P_2-P_1}\left[P_2u_C(0)-\frac{i_L(0)}{C}\right] \qquad A_2=\frac{1}{P_1-P_2}\left[P_1u_C(0)-\frac{i_L(0)}{C}\right]$$

由于P_1、P_2是不相同的负实数，故它们可表示为：

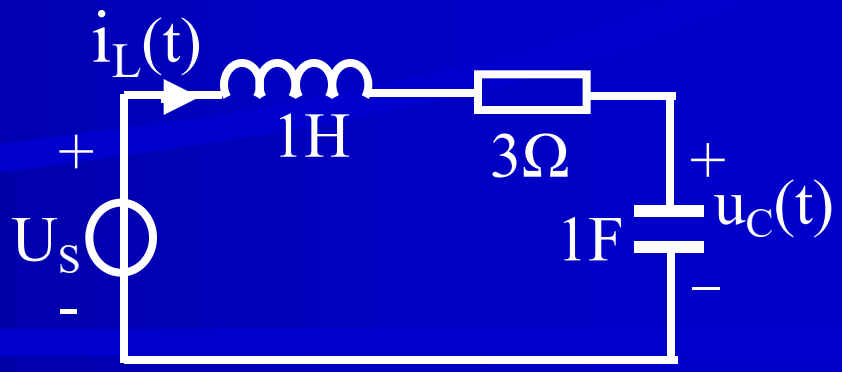
$P_1=-\alpha_1$，$P_2=-\alpha_2$，α小衰减慢，α大衰减快。

$$\therefore u_C(t)=A_1e^{-\alpha_1t}+A_2e^{-\alpha_2t}$$

∴电路的响应是非振荡性的。

例11：已知 $u_c(0)=0$，$i_L(0)=1A$，

$U_S=0$，求$u_c(t)$，$i_L(t)$，$t\geq 0$。



解：特征方程根为：

$$P_{1,2}=-\frac{R}{2L}\pm\sqrt{\left(\frac{R}{2L}\right)^2-\frac{1}{LC}}=-1.5\pm\sqrt{(1.5)^2-1}$$

$$P_1=-0.382\,,\quad P_2=-2.618$$

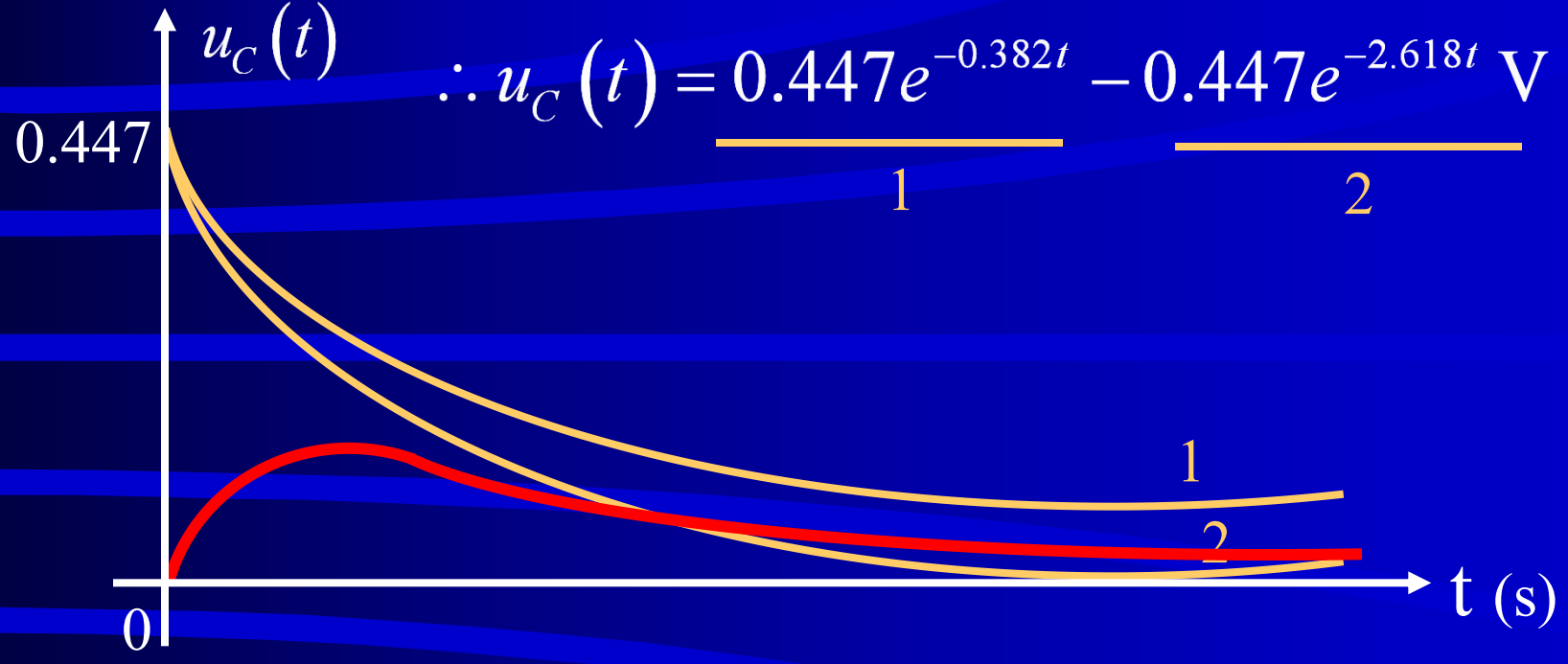
∴微分方程的解：$u_C(t)=A_1e^{P_1t}+A_2e^{P_2t}$

A_1, A_2由初始条件$u_C(0)$, $u_C'(0)$定出。

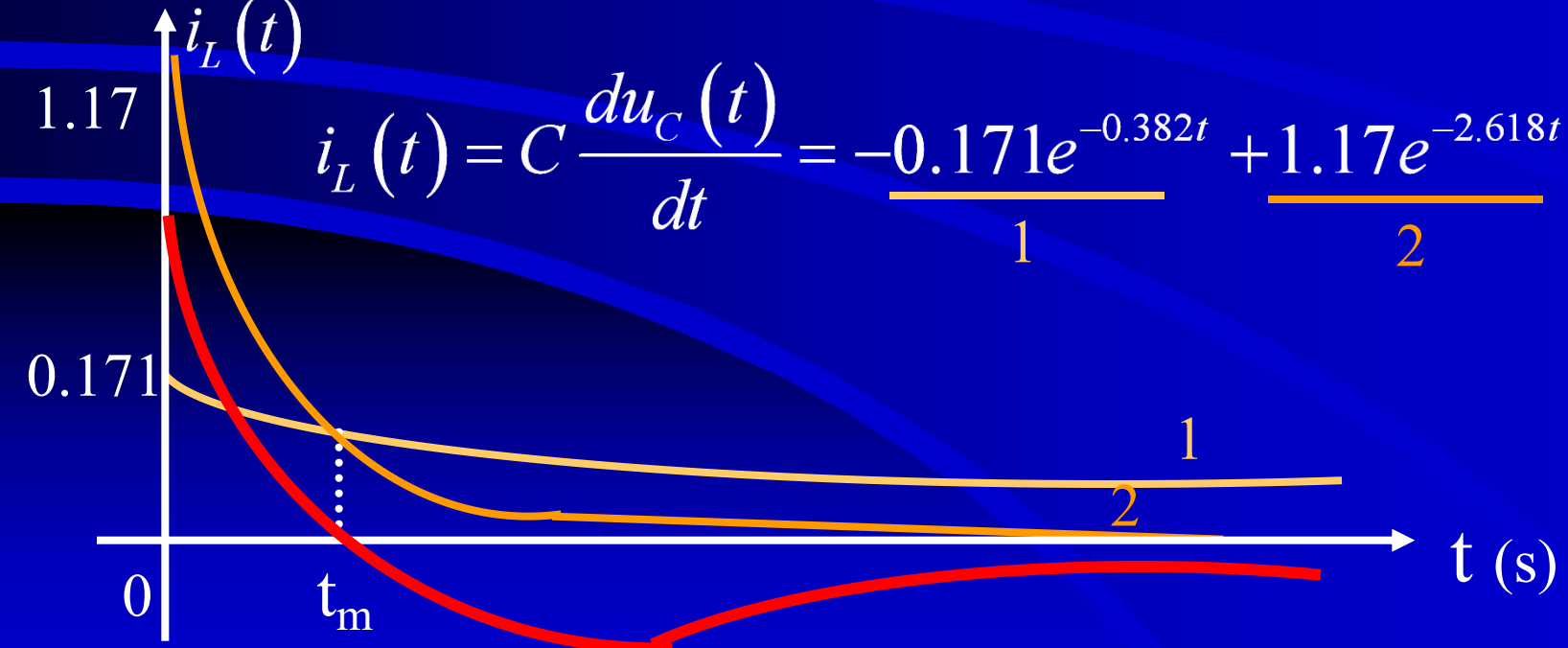
$$\begin{cases} u_C(0)=A_1+A_2=0 \\ u_C'(0)=A_1P_1+A_2P_2=\dfrac{i_L(0)}{C}=\dfrac{1}{1}=1 \end{cases}$$

$$\therefore A_1=0.447$$

$$A_2=-0.447$$



$$\therefore u_C(t) = \underbrace{0.447e^{-0.382t}}_{1} - \underbrace{0.447e^{-2.618t}}_{2}\ \text{V} \quad t \geq 0$$



$$i_L(t) = C\frac{du_C(t)}{dt} = \underbrace{-0.171e^{-0.382t}}_{1} + \underbrace{1.17e^{-2.618t}}_{2}\ \text{A} \quad t \geq 0$$

2. 临界阻尼情况

当$R=2\sqrt{\frac{L}{C}}$时，P_1、P_2为相同的负实数；　　其中：

齐次方程解：　$u_C(t)=(A_1+A_2t)e^{Pt}$　　$P=-\frac{R}{2L}=-\alpha$

A_1, A_2由初始条件定：
$$\begin{cases} u_C(0)=A_1 \\ u_C'(0)=A_1P+A_2=\dfrac{i_L(0)}{C} \end{cases}$$

$$\therefore A_1=u_C(0) \qquad A_2=\frac{i_L(t)}{C}+\alpha u_C(0)$$

$$\therefore u_C(t)=u_C(0)e^{-\alpha t}+\left[\frac{i_L(0)}{C}+\alpha u_C(0)\right]te^{-\alpha t}$$

$$=u_C(0)(1+\alpha t)e^{-\alpha t}+\frac{i_L(0)}{C}te^{-\alpha t}$$

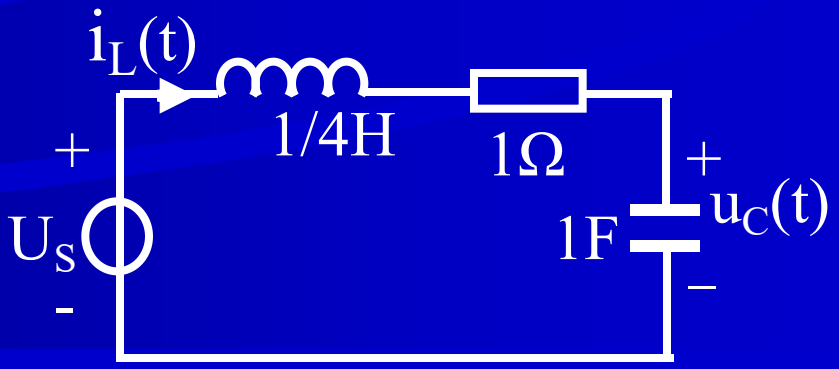
$$\therefore i_L(t)=C\frac{du_C(t)}{dt}$$

$$=-u_C(0)\alpha^2Cte^{-\alpha t}+i_L(0)(1-\alpha t)e^{-\alpha t}$$

∴电路的响应依然是非振荡性的。

例12： 已知 $u_c(0)=-1V$，$i_L(0)=0$，

$U_S=0$，求$i_L(t)$，$t\geq 0$。



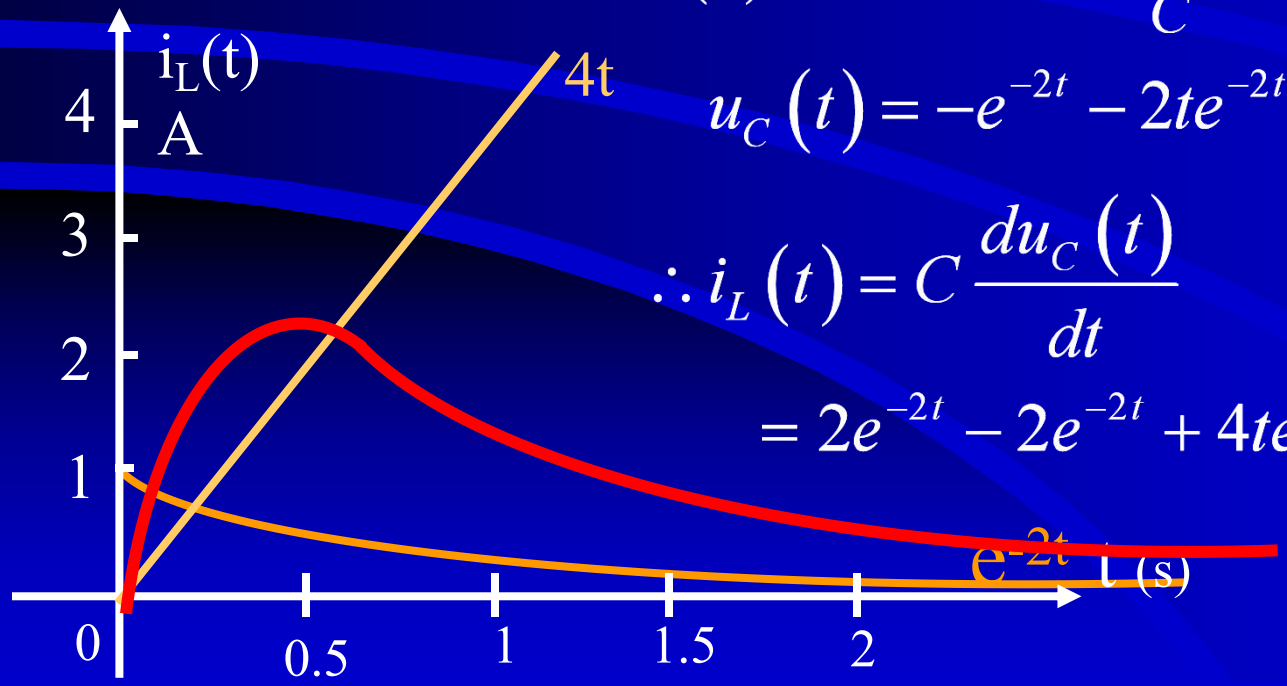
解： 电路的固有频率为：

$$P_{1,2}=-\frac{R}{2L}\pm\sqrt{\left(\frac{R}{2L}\right)^2-\frac{1}{LC}}=-2$$

$\therefore$ 解为： $u_C(t)=(A_1+A_2t)e^{Pt}$

A_1, A_2由初始条件
$$\begin{cases} u_C(0)=A_1=-1 \\ u_C'(0)=A_1P+A_2=\dfrac{i_L(0)}{C}=0 \end{cases}$$

$\therefore A_1=-1,\quad A_2=-2$



$$u_C(t)=-e^{-2t}-2te^{-2t}\ \text{V}\quad t\geq 0$$

$$\therefore i_L(t)=C\frac{du_C(t)}{dt}$$

$$=2e^{-2t}-2e^{-2t}+4te^{-2t}=4te^{-2t}\ \text{A}\quad t\geq 0$$

3. 欠阻尼情况

当$R<2\sqrt{\frac{L}{C}}$时：　P_1、P_2为共轭复数

$$P_{1,2}=-\frac{R}{2L}\pm\sqrt{\left(\frac{R}{2L}\right)^2-\frac{1}{LC}}=-\alpha\pm j\omega_d$$

其中：

$\alpha=\frac{R}{2L}$　衰减系数　α越大衰减越快

$\omega_d=\sqrt{\frac{1}{LC}-\left(\frac{R}{2L}\right)^2}$　衰减振荡角频率

$\omega_0=\frac{1}{\sqrt{LC}}$　振荡时的角频率

齐次方程解：$u_C(t)=e^{-\alpha t}\left(A_1Cos\omega_d t+A_2Sin\omega_d t\right)$

齐次方程解：$u_C(t)=e^{-\alpha t}\left(A_1 Cos\omega_d t+A_2 Sin\omega_d t\right)$

A_1, A_2 由初始条件定：

$$\begin{cases} u_C(0)=A_1 \\ u_C'(0)=-\alpha A_1+\omega_d A_2=\dfrac{i_L(0)}{C} \end{cases}$$

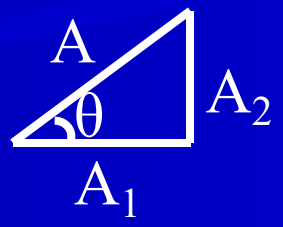
$$\therefore A_1=u_C(0) \quad A_2=\frac{1}{\omega_d}\left[\alpha u_C(0)+\frac{i_L(0)}{C}\right]$$

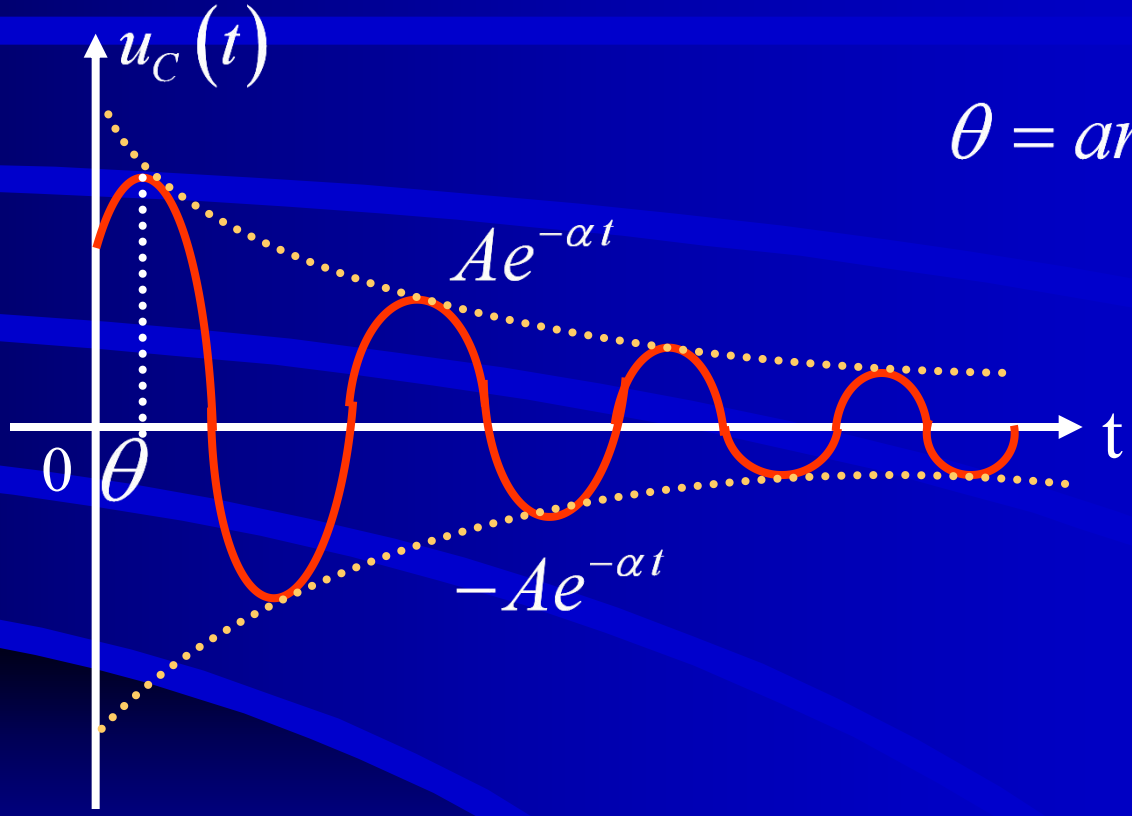
$$u_C(t)=\underbrace{e^{-\alpha t}\sqrt{A_1^2+A_2^2}}_{A}\left(\underbrace{\frac{A_1}{\sqrt{A_1^2+A_2^2}}}_{Cos\theta} Cos\omega_d t+\underbrace{\frac{A_2}{\sqrt{A_1^2+A_2^2}}}_{Sin\theta} Sin\omega_d t\right)$$

$$\because Cos(\alpha\pm\beta)=Cos\alpha\, Cos\beta\mp Sin\alpha\, Sin\beta$$

$$\therefore u_C(t)=Ae^{-\alpha t}Cos(\omega_d t-\theta)$$

$$\therefore u_C(t) = Ae^{-\alpha t} Cos(\omega_d t - \theta)$$

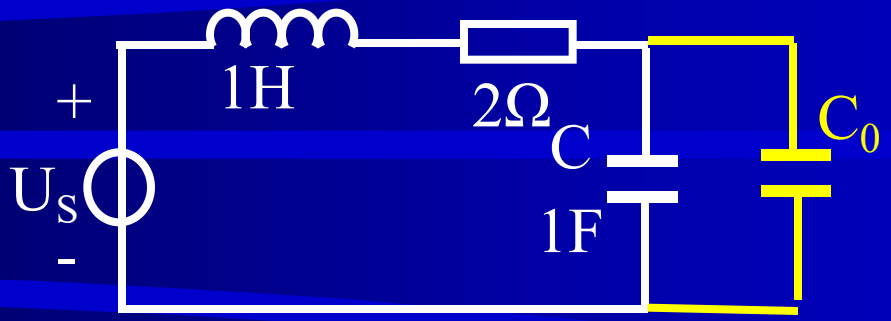




$$\theta = arc\,tg\,\frac{A_2}{A_1}$$

例13:求在原电路增加一个元件后，电路的状态。

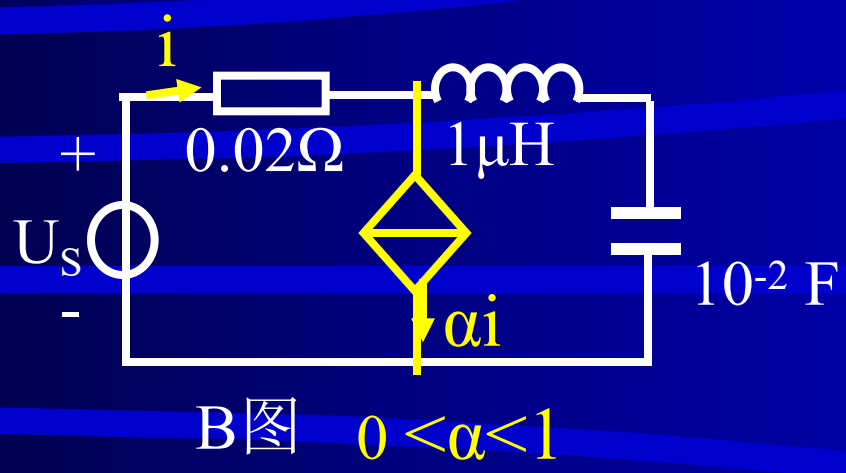
解:



A图

$$R = 2\sqrt{\frac{L}{C}}$$ 临界阻尼

加C_0后： $$R > 2\sqrt{\frac{L}{C + C_0}}$$ 过阻尼



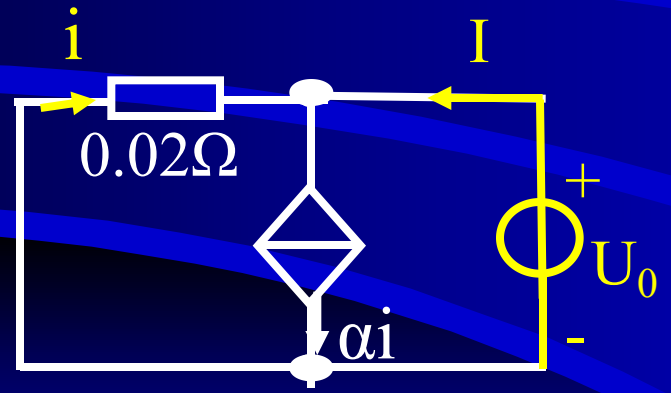
B图 $0<\alpha<1$

$$R=2\sqrt{\frac{L}{C}}$$

$$\therefore 0.02=2\sqrt{\frac{10^{-6}}{10^{-2}}}$$

临界阻尼

加受控源：



$0<\alpha<1$

用外加电压法求$R_0=U_0/I$

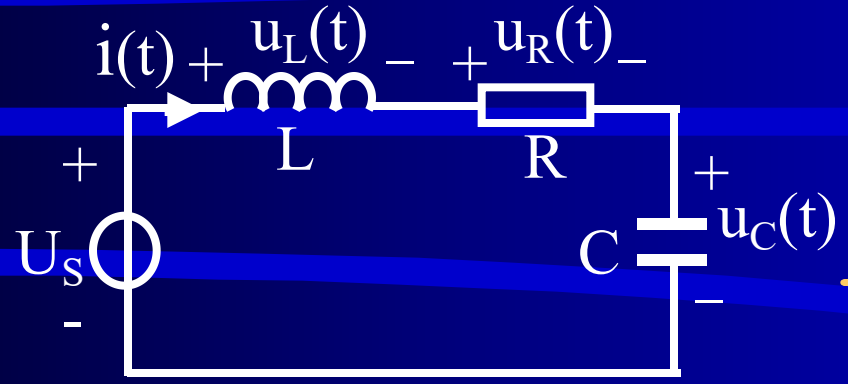
$I+i=\alpha i$ $\quad\therefore I=(\alpha-1)i$

又：$U_0=-0.02\,i$

$$\therefore R_0=\frac{U_0}{I}=\frac{0.02}{1-\alpha}>R$$

∴电路从临界阻尼→过阻尼

5.4.2 二阶电路的零状态响应与完全响应



由KVL: $u_L + u_R + u_C = U_S$

$$LC\frac{d^2u_C}{dt^2} + RC\frac{du_C}{dt} + u_C = U_S$$

$$\begin{cases} u_C(0) = U_0 \\ u_C'(t)_{t=0} = \dfrac{i_L(0)}{C} \end{cases}$$

解：$u_c(t) = u_{ch}(t) + u_{cp}(t)$

通解：1. $u_{Ch}(t) = A_1 e^{P_1 t} + A_2 e^{P_2 t}$ 过阻尼

2. $u_{Ch}(t) = (A_1 + A_2 t)e^{Pt}$ 临界阻尼

3. $u_{Ch}(t) = e^{-\alpha t}\left(A_1 Cos\omega_d t + A_2 Sin\omega_d t\right)$ 欠阻尼

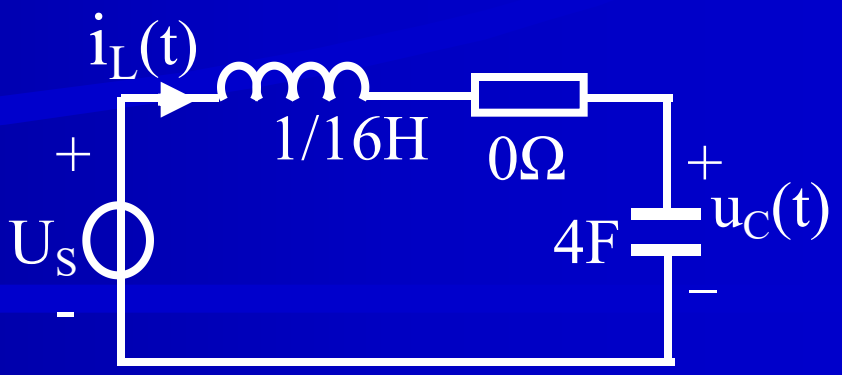
特解：$u_{Cp}(t) = U_S$

∴ 完全解：$u_C(t) = A_1 e^{-\alpha_1 t} + A_2 e^{-\alpha_2 t} + U_S \quad t \geq 0$

A_1,A_2由$u_c(0)$,$i_L(0)$定出。

例14：已知 $u_c(0)=-5V$，$i_L(0)=0$，

$U_S=10V$，求$u_c(t)$，$i_L(t)$, $t\geq 0$。



解：电路的微分方程为：

$$LC\frac{d^2u_C}{dt^2}+u_C=U_S$$

特征方程为：$LCP^2+1=0$

特征方程根为：$P_{1,2}=\pm j\frac{1}{\sqrt{LC}}=\pm j\omega_0=\pm j2$

通解：$u_{Ch}(t)=A_1Cos\omega_0 t+A_2Sin\omega_0 t$

特解：$u_{Cp}(t)=U_S=10V$

完全解：$u_C(t)=A_1Cos\omega_0 t+A_2Sin\omega_0 t+10$

A_1, A_2由初始条件 $\begin{cases} u_C(0) = A_1 + 10 = -5 \\ u_C'(t)_{t=0} = \omega_0 A_2 = \dfrac{i_L(0)}{C} = 0 \end{cases}$

解得: $A_1 = -15, \quad A_2 = 0$

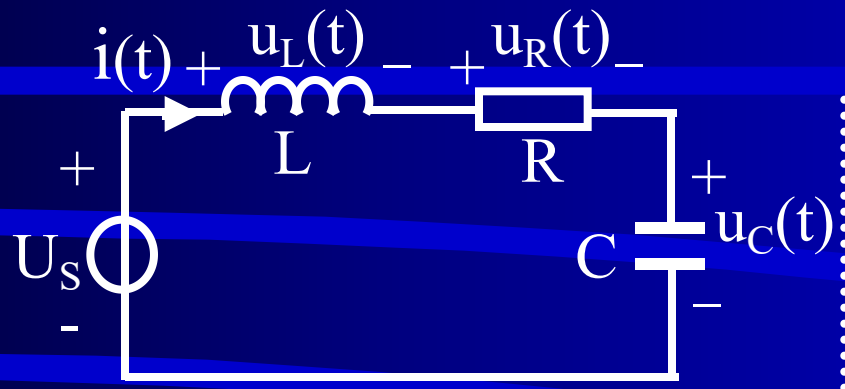
$$\therefore u_C(t) = -15Cos2t + 10 \quad t \geq 0$$

$$\therefore i_L(t) = C\frac{du_C(t)}{dt} = 4\frac{d}{dt}(-15Cos2t + 10)$$

$$= 120Sin2t \quad t \geq 0$$

5.4.3 并联二阶电路的分析

—与串联电路分析是对偶的



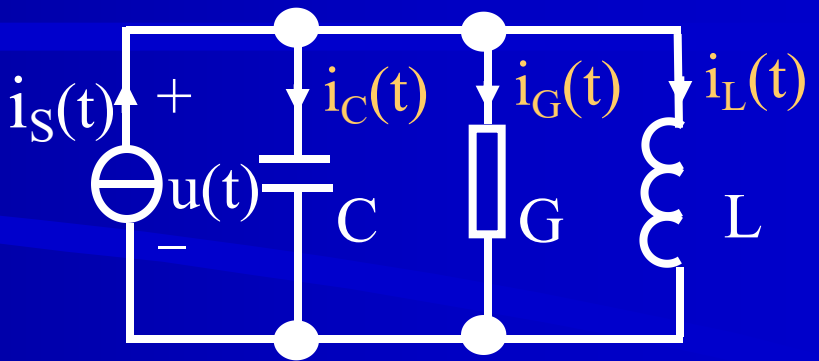
选u_C为求解变量

由KVL: $u_L + u_R + u_C = U_S$

$$LC\frac{d^2u_C}{dt^2} + RC\frac{du_C}{dt} + u_C = U_S$$

$$LC\,P^2 + RC\,P + 1 = 0$$

$$P_{1,2} = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$



选i_L为求解变量

由KCL: $i_L + i_G + i_C = i_S$

$$LC\frac{d^2i_L}{dt^2} + GL\frac{di_L}{dt} + i_L = i_S$$

$$LC\,P^2 + GL\,P + 1 = 0$$

$$P_{1,2} = -\frac{G}{2C} \pm \sqrt{\left(\frac{G}{2C}\right)^2 - \frac{1}{LC}}$$

1. $R > 2\sqrt{\frac{L}{C}}$，过阻尼，非振荡

P_1、P_2为不相同的负实数；

$$u_C(t) = A_1 e^{p_1 t} + A_2 e^{p_2 t}$$

2. $R = 2\sqrt{\frac{L}{C}}$，临界阻尼，非振荡

P_1、P_2为相同的负实数；

$$u_C(t) = (A_1 + A_2 t) e^{Pt}$$

3. $R < 2\sqrt{\frac{L}{C}}$，欠阻尼，衰减振荡

P_1、P_2为共轭复数；

$$u_C(t) = e^{-\alpha t}\left(A_1 Cos\omega_d t + A_2 Sin\omega_d t\right)$$

1. $G > 2\sqrt{\frac{C}{L}}$，过阻尼，非振荡

P_1、P_2为不相同的负实数；

$$i_L(t) = A_1 e^{P_1 t} + A_2 e^{P_2 t}$$

2. $G = 2\sqrt{\frac{C}{L}}$，临界阻尼，非振荡

P_1、P_2为相同的负实数；

$$i_L(t) = (A_1 + A_2 t) e^{Pt}$$

3. $G < 2\sqrt{\frac{C}{L}}$，欠阻尼，衰减振荡

P_1、P_2为共轭复数；

$$i_L(t) = e^{-\alpha t}\left(A_1 Cos\omega_d t + A_2 Sin\omega_d t\right)$$

4. $R=0$， 等幅振荡

P_1、P_2为虚数；（无阻尼）

$$u_C(t)=A_1Cos\omega_0 t+A_2Sin\omega_0 t$$

5.由初始条件定常数A：

$$\begin{cases} u_C(0)=U_0 \\ u_C'(t)_{t=0}=\left.\dfrac{du_C(t)}{dt}\right|_{t=0}=\dfrac{i_L(0)}{C} \end{cases}$$

$\because LC$串联，$i_L=i_C=C\dfrac{du_C(t)}{dt}$

$$\therefore \left.\frac{du_C(t)}{dt}\right|_{t=0}=\frac{i_L(0)}{C}$$

4. $G=0$， 等幅振荡

P_1、P_2为虚数；（无阻尼）

$$i_L(t)=A_1Cos\omega_0 t+A_2Sin\omega_0 t$$

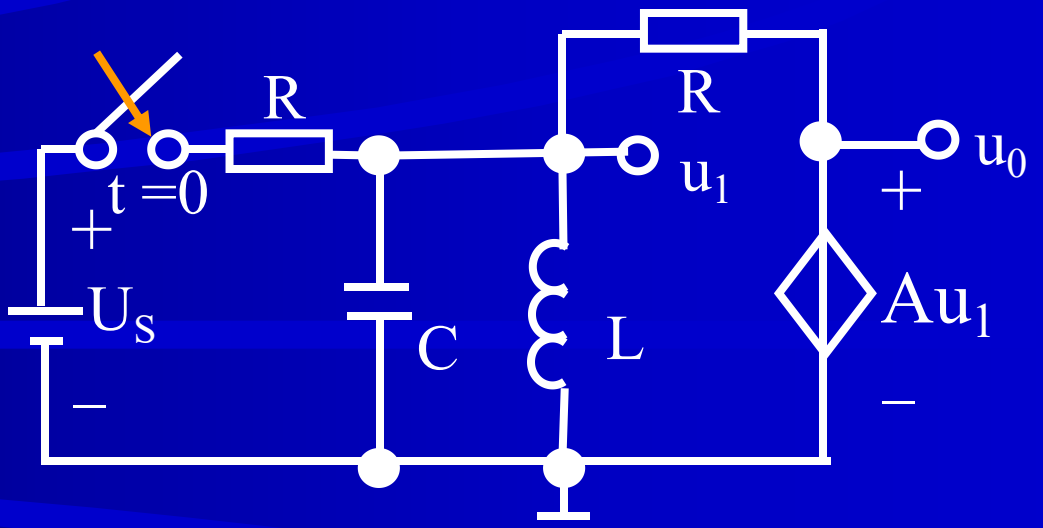
5.由初始条件定常数A：

$$\begin{cases} i_L(0)=I_0 \\ i_L'(t)_{t=0}=\left.\dfrac{di_L(t)}{dt}\right|_{t=0}=\dfrac{u_C(0)}{L} \end{cases}$$

$\because LC$并联，$u_C=u_L=L\dfrac{di_L(t)}{dt}$

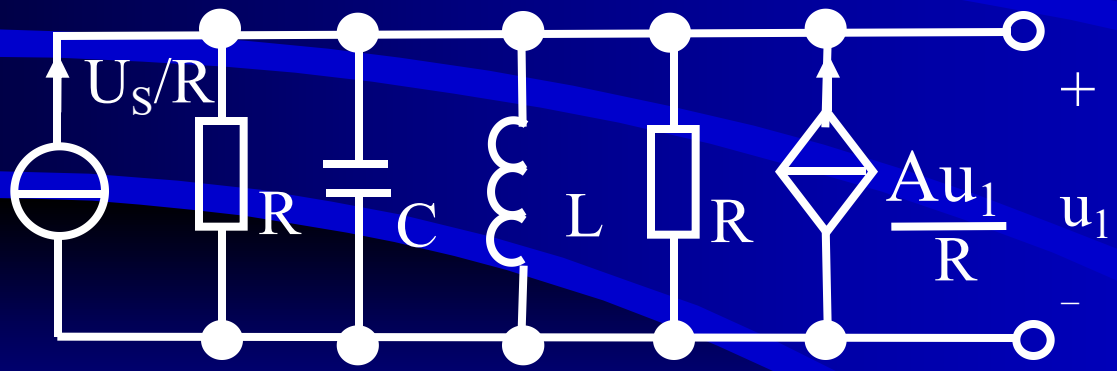
$$\therefore \left.\frac{di_L(t)}{dt}\right|_{t=0}=\frac{u_C(0)}{L}$$

例15：　若u_0为等幅振荡时,A=?.已知初始状态为零,求 $u_0(t)$=? , t≥0。



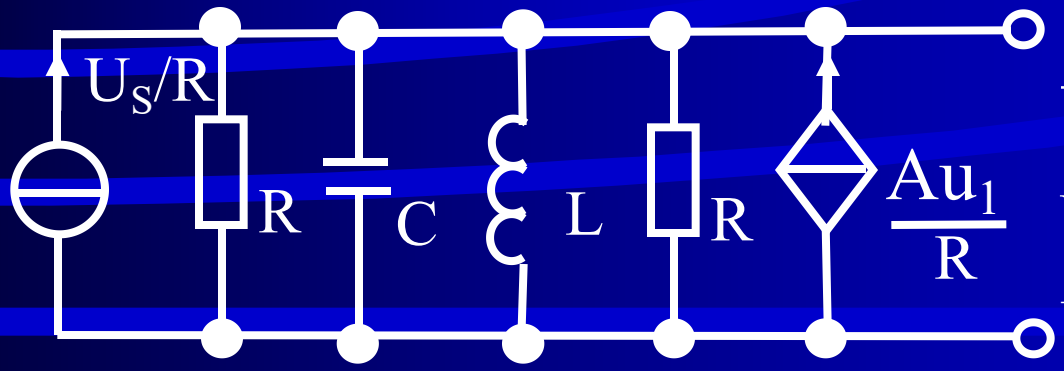
解：　t >0时:

已知$U_0=Au_1$,∴ u_0为等幅振荡时, u_1也为等幅振荡.



选u_1为求解变量

由KCL:　$$i_L + i_R + i_C = \frac{U_S}{R} + \frac{Au_1}{R}$$



$$i_L + i_R + i_C = \frac{U_S}{R} + \frac{Au_1}{R}$$

$$\therefore \frac{2}{R}u_1 + C\frac{du_1}{dt} + \frac{1}{L}\int_{-\infty}^{t} u_1 dt = \frac{U_S}{R} + \frac{Au_1}{R}$$

对u_1求一次导,整理后:

$$\frac{d^2u_1}{dt^2} + \frac{1}{RC}(2-A)\frac{du_1}{dt} + \frac{1}{LC}u_1 = 0$$

$\therefore$ 取$A=2$,微分方程为: $\dfrac{d^2u_1}{dt^2} + \dfrac{1}{LC}u_1 = 0$

特征方程为：$P^2+\frac{1}{LC}=0$

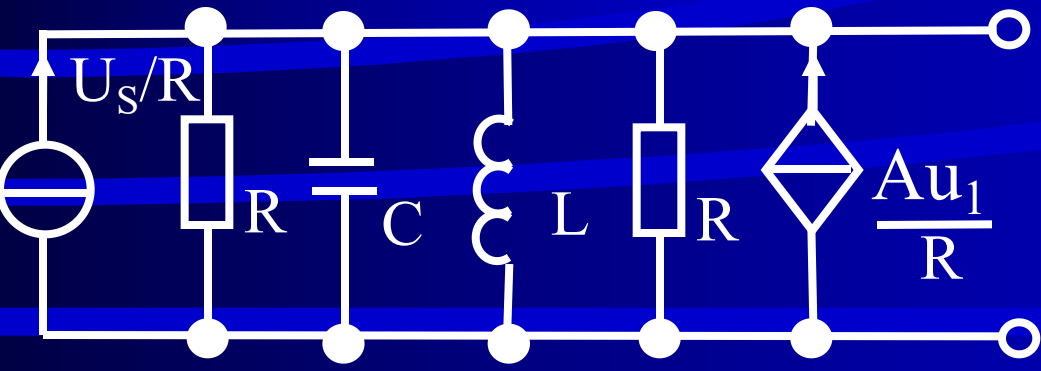
特征方程根为：$P_{1,2}=\pm j\frac{1}{\sqrt{LC}}$

$$u_1(t)=A_1Cos\omega_0 t+A_2Sin\omega_0 t \quad t\geq 0$$

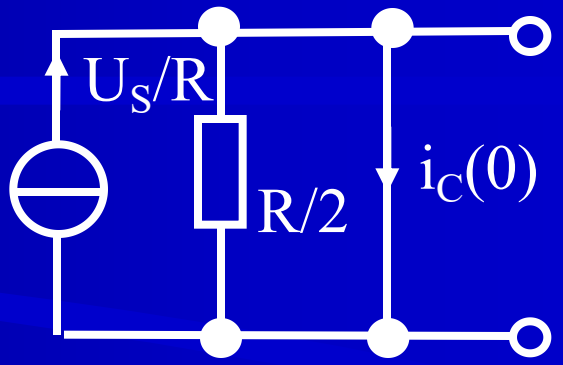
由初始条件定 A_1, A_2

$$\begin{cases} u_1(0)=A_1=0 \\ u_1'(0)=\omega_0 A_2=\dfrac{i_C(0)}{C} \end{cases}$$

∵LC是并联，$i_C(0)\neq i_L(0)$，∴$i_C(0)$要在$t=0_+$的等效电路中求出。



$t=0_+$ 时，C→ 短路，L→开路。



$$i_C(0)=\frac{U_S}{R}$$

$$\begin{cases} u_1(0)=A_1=0 \\ u_1'(0)=\omega_0 A_2=\dfrac{i_C(0)}{C}=\dfrac{U_S/R}{C} \end{cases} \Rightarrow \begin{cases} A_1=0 \\ A_2=\sqrt{\dfrac{L}{C}}\dfrac{U_S}{R} \end{cases}$$

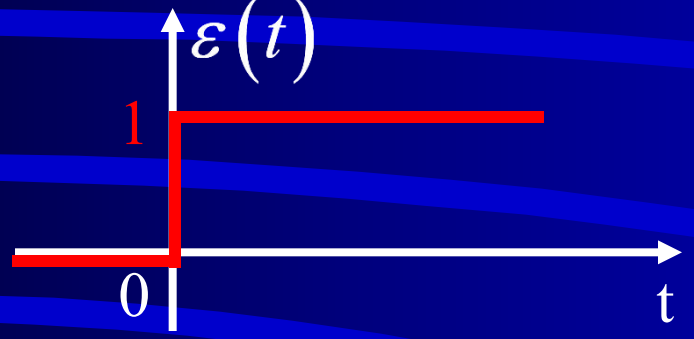
$$\therefore u_1(t)=A_2 Sin\omega_0 t=\frac{U_S}{R}\sqrt{\frac{L}{C}}Sin\frac{1}{\sqrt{LC}}t \quad t\geq 0$$

$$\therefore u_0(t)=Au_1(t)=2\frac{U_S}{R}\sqrt{\frac{L}{C}}Sin\frac{1}{\sqrt{LC}}t \quad t\geq 0$$ ※

5.5 阶跃函数和阶跃响应

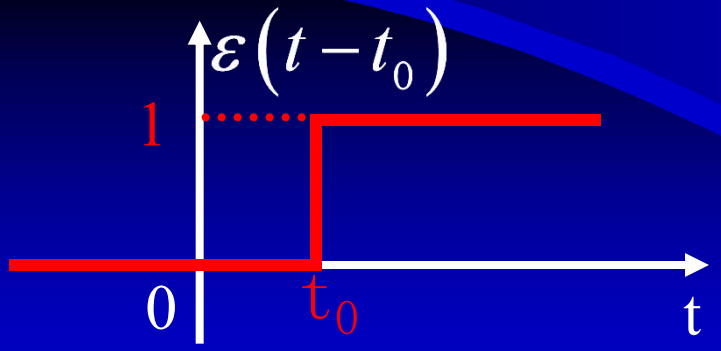
5.5.1 阶跃函数

1. 单位阶跃函数



$$\varepsilon(t)=\begin{cases}1 & t>0\\ 0 & t<0\end{cases}$$

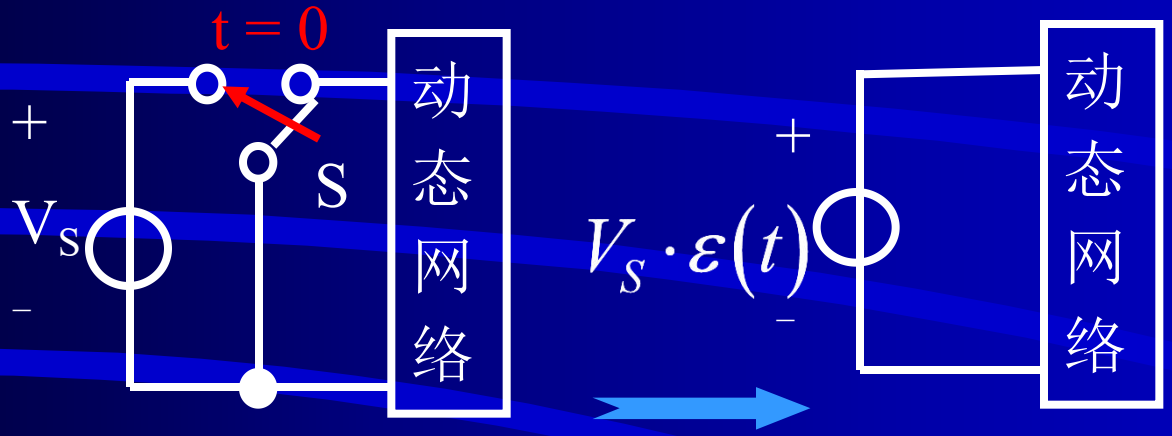
2. 延时单位阶跃函数

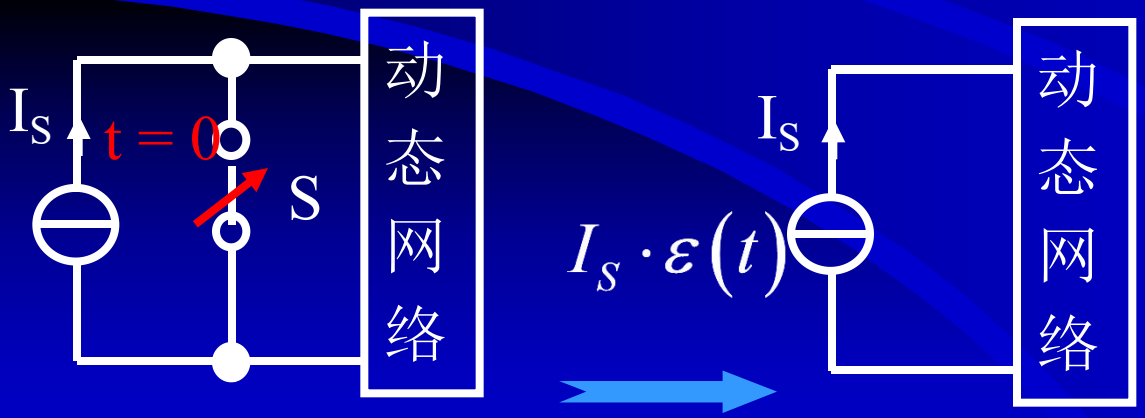


$$\varepsilon(t-t_0)=\begin{cases}1 & t>t_0\\ 0 & t<t_0\end{cases}$$

5.5.2 阶跃函数的作用

1. 模拟开关的作用





$$\because \varepsilon(t)=\begin{cases}1 & t>0\\ 0 & t<0\end{cases}$$

$\therefore t<0$时：

$V_S \cdot \varepsilon(t)=0$

$I_S \cdot \varepsilon(t)=0$

V_S, I_S不起作用

$\therefore t>0$时：

$V_S \cdot \varepsilon(t)=V_S$

$I_S \cdot \varepsilon(t)=I_S$

V_S, I_S作用

2. 表示时间区间

如果某一电路的解为

$$u_c(t) = 4\left(1 - e^{-2t}\right)\text{V} \qquad t \geq 0$$

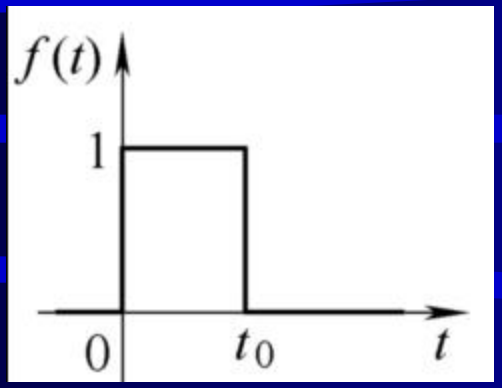
$u_c(t) = 4(1+e^{-2t})\text{V}$

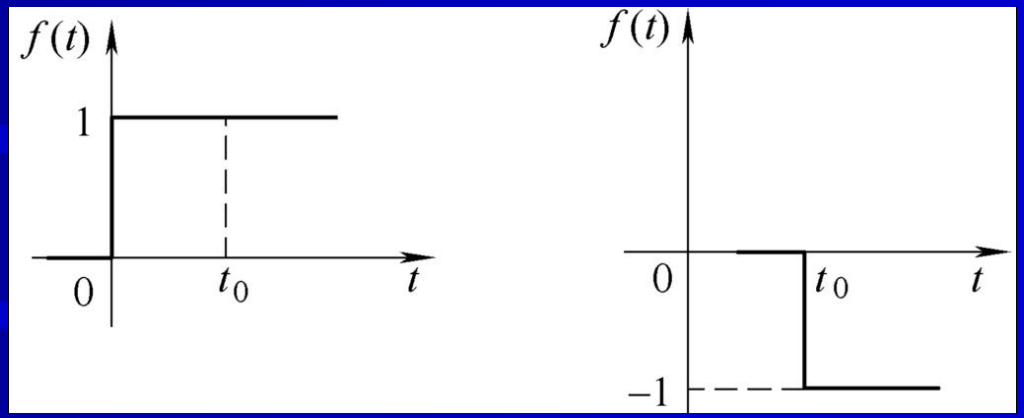
$t \geq 0$

用阶跃函数可以表示为

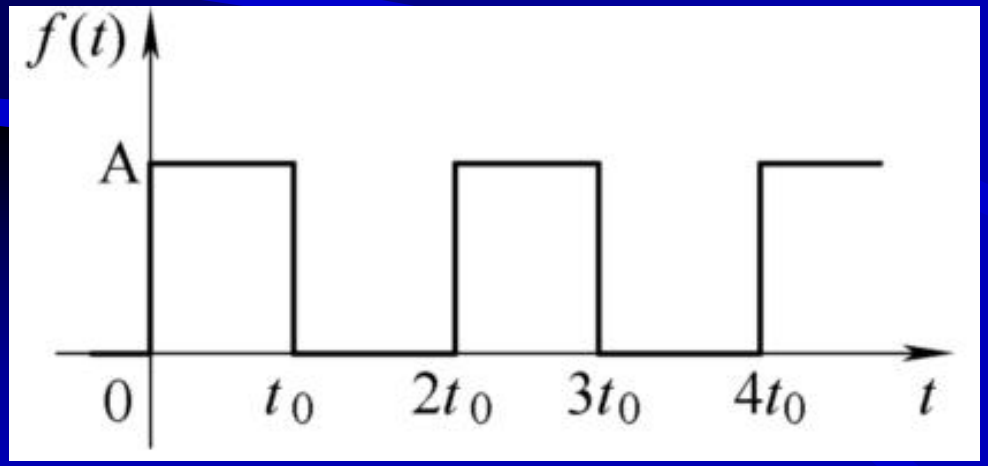
$$u_c(t) = 4\left(1 - e^{-2t}\right)\varepsilon(t)\ \text{V}$$

3.表示分段常数信号





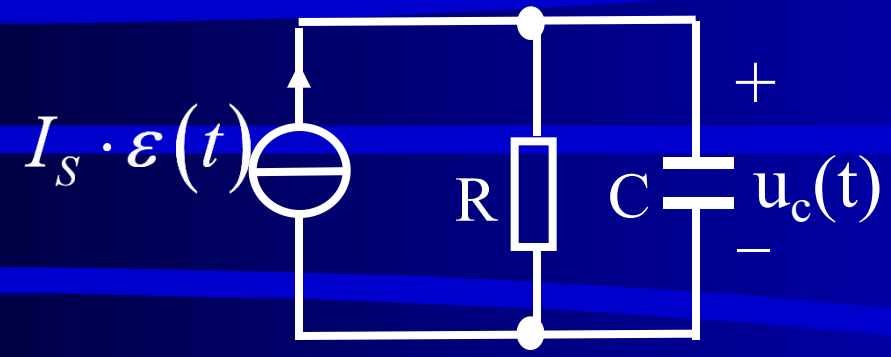
$$f(t)=\varepsilon(t)-\varepsilon(t-t_0)$$



$$f(t)=A\varepsilon(t)-A\varepsilon(t-t_0)+A\varepsilon(t-2t_0)-A\varepsilon(t-3t_0)+....$$

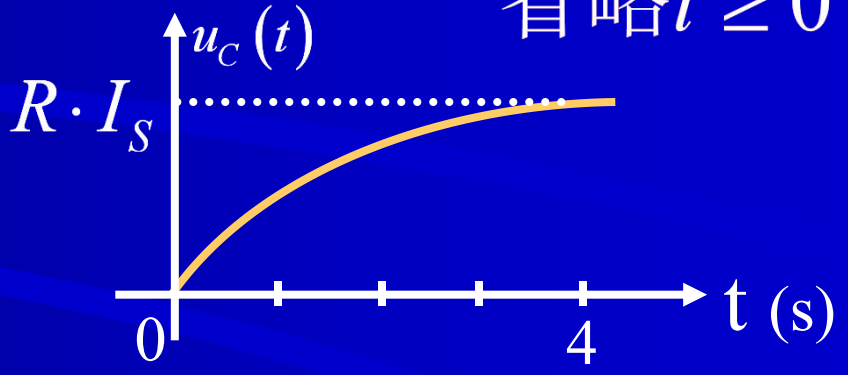
4. 阶跃函数响应

零状态响应：



$$u_C(t) = R \cdot I_S \left(1 - e^{-\frac{t}{\tau}}\right) \varepsilon(t)$$

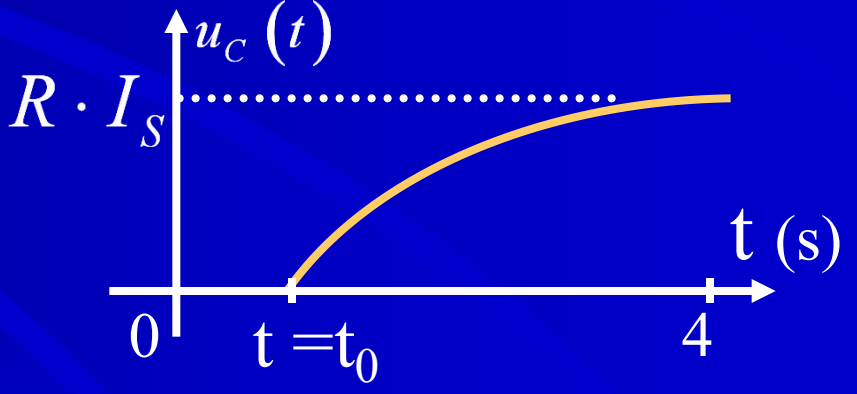
省略$t \geq 0$



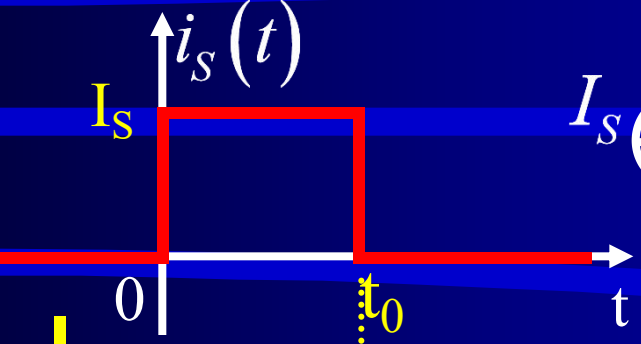
若电流源在$t \geq t_0$时作用与电路，激励为$I_S \cdot \varepsilon(t - t_0)$

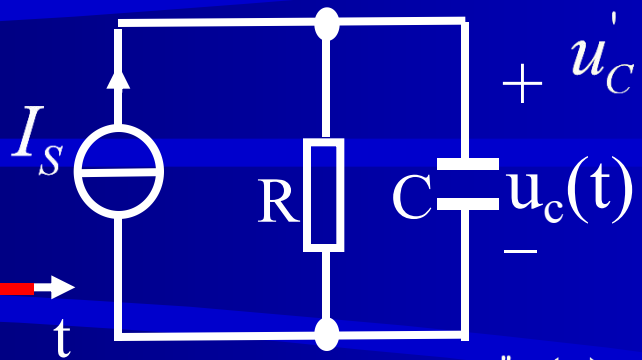
$$u_C(t) = R \cdot I_S \left(1 - e^{-\frac{t}{\tau}}\right) \varepsilon(t - t_0)$$

省略$t \geq t_0$



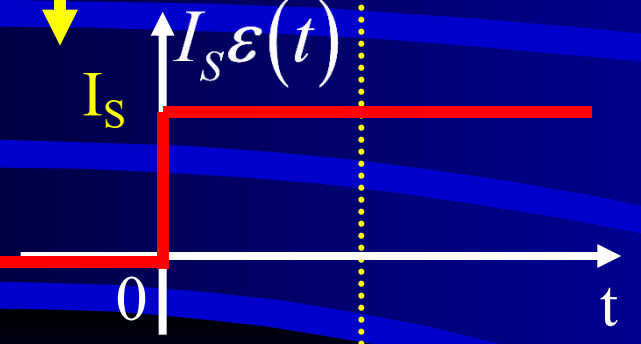
单位阶跃响应

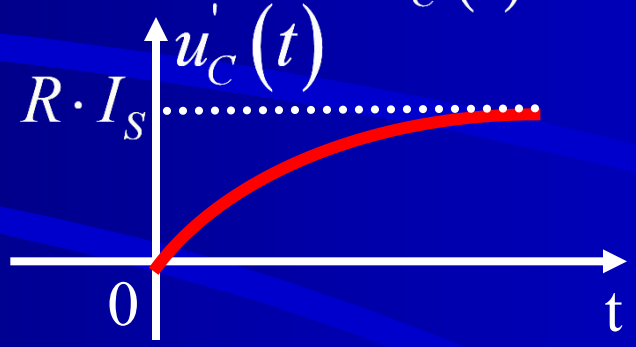


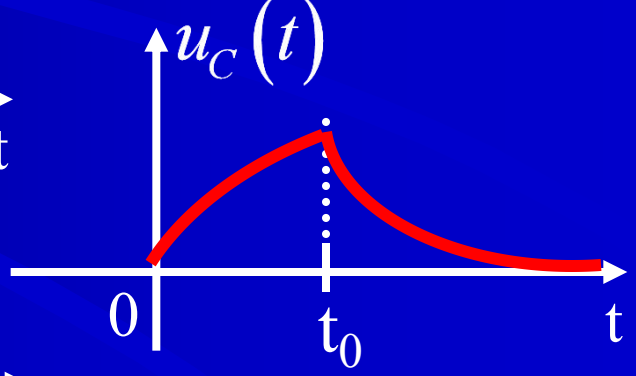


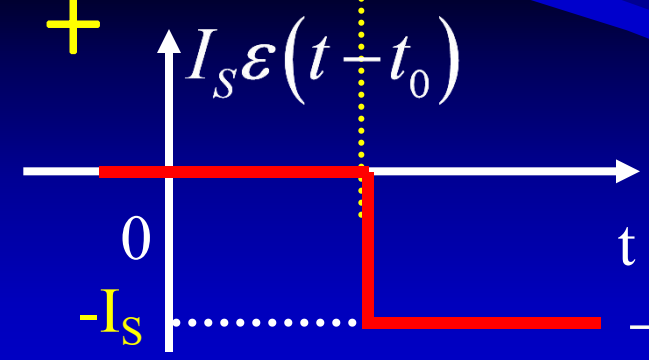
$$u_C'(t) = R \cdot I_S \left(1 - e^{-\frac{t}{\tau}}\right) \varepsilon(t)$$

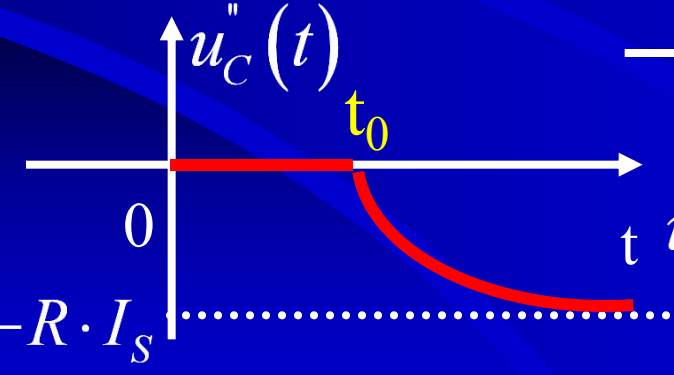
$$u_C''(t) = -R \cdot I_S \left(1 - e^{-\frac{t-t_0}{\tau}}\right) \varepsilon(t-t_0)$$





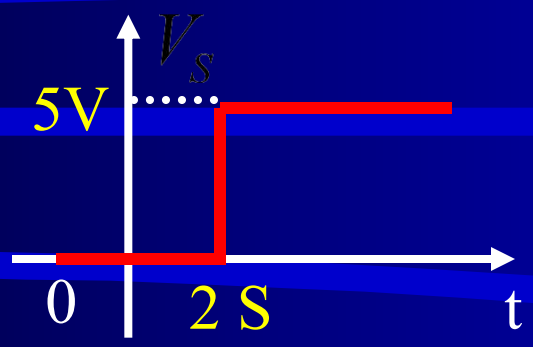


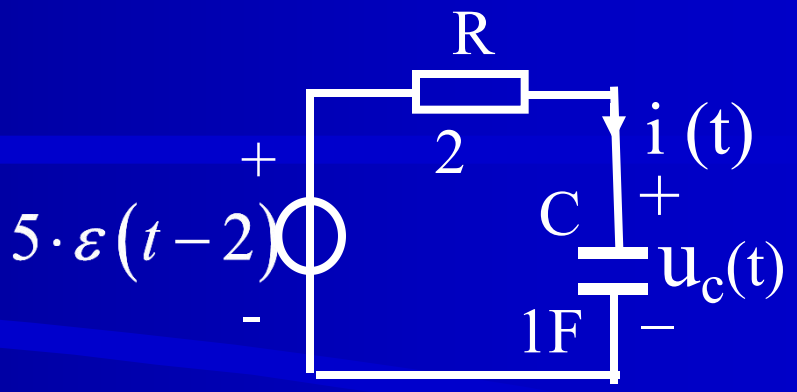




$$u_C(t) = u_C'(t) + u_C''(t)$$

例16： 已知 V_S如下图， $u_c(0)=10$,求 i (t)。





$$\tau = R\cdot C = 2\text{S}$$

解： 1. t = 0时， $u_c(0)=10$V, （零输入响应）

$$\therefore u_C(t) = u_C(0)e^{-\frac{t}{\tau}}\cdot\varepsilon(t) = 10e^{-\frac{t}{2}}\cdot\varepsilon(t)$$

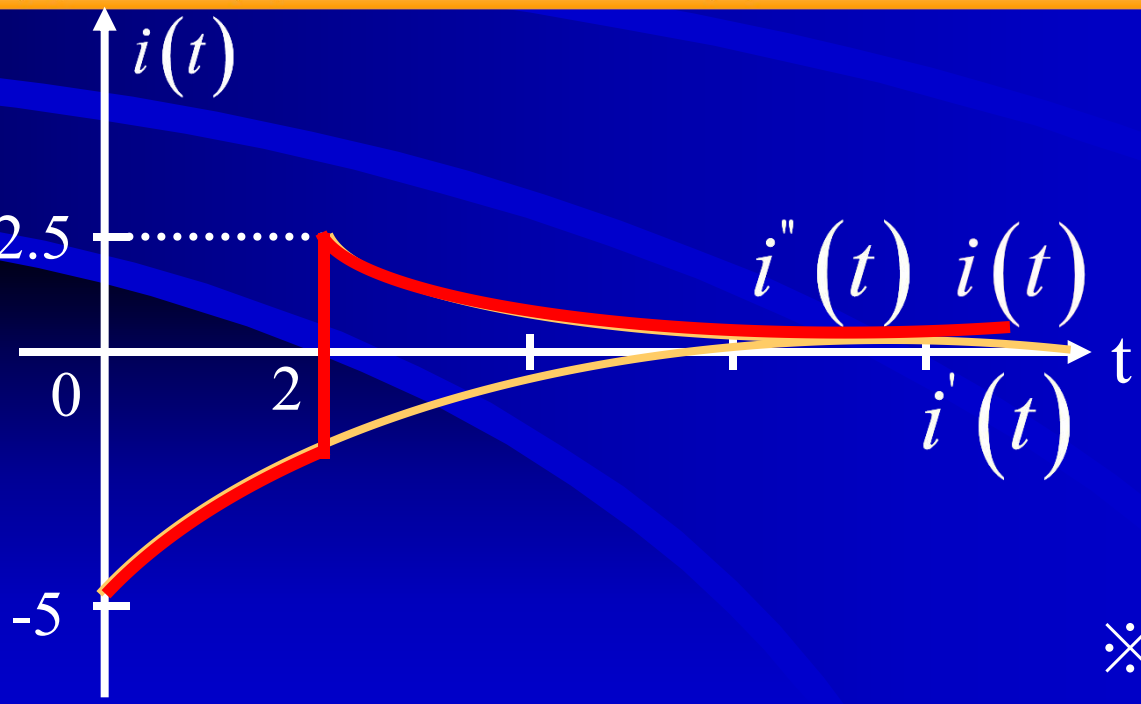
$$i'(t) = C\frac{du_C(t)}{dt} = -5e^{-\frac{t}{2}}\cdot\varepsilon(t)$$

2. t > 2S时，（零状态响应）

$$\therefore u_C(t) = V_S\left(1-e^{-\frac{t}{\tau}}\right)\cdot\varepsilon(t-2) = 5\left(1-e^{-\frac{t-2}{2}}\right)\cdot\varepsilon(t-2)$$

$$i''(t) = C\frac{du_C(t)}{dt} = 2.5e^{-\frac{t-2}{2}}\cdot\varepsilon(t-2)$$

$$\therefore i(t) = i'(t)+i''(t) = -5e^{-\frac{t}{2}}\cdot\varepsilon(t)+2.5e^{-\frac{t-2}{2}}\cdot\varepsilon(t-2)$$



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谢谢大家！

再见

再见！

